Until we are all equal

Participatory Climate Risk and Vulnerability Assessment for Gender Responsive Climate Adaptation and Preparedness Plans (GRAPP) in Kilifi County

Draft Sokoke Ward Adaptation Plan

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TABLE OF CONTENTS

List of Figures iii
List of Tables iii

1 Chapter 1: Background Information 1
1.1 Introduction 1
1.2 Plan development approach 2
1.2.1 Inception Phase 3
1.2.2 Participatory Climate Risk and Vulnerability Assessment (PCVA) 4
1.2.3 Household Survey 9
1.2.4 Forest Health Assessment 9
1.2.5 Current and Future Climate Projections 10
1.2.6 Data Analysis and Reporting 10
1.2.7 Validation 11

2 Chapter 2: Coordination of Adaptation Actions 12
2.1 Introduction 12
2.2 Current Institutional Arrangements 12
2.3 Relevant Policies 13
2.4 Proposed Coordination Arrangements 13
2.5 Implementation Roles 14

3 Chapter 3: Situation analysis of climate vulnerability 16
3.1 Introduction 16
3.2 Current climate trends 16
3.3 Future climate trends 17
3.4 Climate hazards and vulnerability 18
3.4.1 Drought 18
3.4.2 Human –wildlife conflicts 21
3.4.3 Environmental Degradation 23
3.4.1 Pests and Diseases 26
3.4.2 Floods 28
3.4.3 Whirl Wind 30
3.5 Classification of Exposure and Resilience through Livelihoods 32
3.5.1 Livelihood Structure and Monthly Incomes 32
3.5.2 Vulnerabilities of Livelihood Options 34
3.5.3 Overall Classification and Implications 36
3.6 Analyzing causes of vulnerability. 36
3.6.1 Classification of Severity of the Vulnerabilities and Their Causes 36
3.6.2 Identification of Unsafe Conditions 36
3.6.3 Identification of Dynamic Pressures or Determinants of Vulnerability 37
3.6.4 Identification of Underlying Causes of Vulnerability 37
3.6.5 Prioritization of Causes 38

4 Chapter 4: Gender/Age Vulnerability Analysis for Climate Hazards 39
4.1 Children's Vulnerability to Climate Hazards and Key Mitigation Measures in Sokoke Ward45

5 Chapter 4: Adaptation Actions 47
5.1 Introduction 47
5.2 Existing climate-resilient and livelihood-enhancing projects 47
5.3 Adaptation actions 50
5.4 Resource Mobilisation 58

6 Chapter 6: Adaptation Monitoring and Evaluation 59
6.1 Introduction 59
6.2 The Adaptation Theory of Change 59
6.3 Performance Indicators for Adaptation 60
6.4 Adaptation Reporting and Learning 64

7 References 67

8 Acknowledgements 69

9 Annexure 70

List of Figures

Figure 1: Location of Sokoke Ward and survey sites Source: PCVA, 2025) 2
Figure 2: A model of factors that can be used to assess the vulnerability and resilience of systems (Source: Füssel and Klein, 2006, and Gauthier et al., 2014) 3
Figure 3: Focus Group Discussions in Sokoke Ward (Source: Field Surveys,2025) 6
Figure 4: Hazard and Resource Mapping in Dzikudze Village, Sokoke Ward (Source: PCVA, 2025 7
Figure 5: SSP2 4.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022) ..17
Figure 6: SSP2 4.5 Mean Temperature and Land Surface Temperature Projections. (Source: Chirps database and ERA5-Ag, 2022) 17
Figure 7: SSP5 8.5 Rainfall, AET and PET Projections. (Source: Source: Chirps database and ERA5-Ag, 2022) 18
Figure 8 : SSP5 8.5 Mean Temperature and Land Surface Temperature Projections. (Source: Source: Chirps database and ERA5-Ag, 2022) 18
Figure 9: Drought prevalence historical trend (Source: PCVA, 2025) 19
Figure 10: Drought vulnerability score in Sokoke Ward (Source: PCVA, 2025) 19
Figure 11: Mwangea Hills NDVI values across dry and wet seasons (Source: Source: Landsat 8/9 OLI and TIRS, 2024) 20
Figure 12: Human-wildlife prevalence historical trend (Source: PCVA, 2025) 22
Figure 13: Human-wildlife vulnerability score in Sokoke Ward (Source: PCVA, 2025) 22
Figure 14: Environmental Degradation prevalence historical trend (Source: PCVA, 2025) 24
Figure 15: Environmental Degradation vulnerability score in Sokoke Ward (Source: PCVA, 2025) 24
Figure 16: Pests and diseases prevalence historical trend (Source: PCVA, 2025) 26
Figure 17: Pests and diseases vulnerability score in Sokoke Ward (Source: PCVA, 2025) 27
Figure 18: Floods prevalence historical trend (Source: PCVA, 2025) 28
Figure 19: Floods vulnerability score in Sokoke Ward (Source: PCVA, 2025) 29
Figure 20: Whirlwinds prevalence historical trend (Source: PCVA, 2025) 31
Figure 21: Whirl Wind vulnerability score in Sokoke Ward (Source: PCVA, 2025) 31
Figure 22: Respondents who lacked money to Meet Basic Needs in the Last 12 Months (Source: PCVA 2025) 34

List of Tables

Table 1: Institutional Roles in Implementation of Adaptation Actions 14
Table 2: Monthly earnings by livelihood and village (Source: PCVA, 2025) 33
Table 3: Gender/Age Vulnerability Analysis for Climate Hazards 42
Table 4: Existing climate-resilient and livelihood-enhancing projects in Sokoke Ward 49
Table 5: Adaptation Actions for Sokoke Ward 53

1 Chapter 1: Background Information

1.1 Introduction

Plan international Kenya is implementing the Conservation and Sustainable Management of Marine and Coastal Ecosystems (COSME) Project in Kilifi and Kwale Counties. The two counties are endowed with rich coastal and marine ecosystems ranging from terrestrial forests, mangroves, seagrass beds, coral reefs, marine species and open sea marine resources among others. These natural assets not only underpin livelihoods but also drive key economic sectors including tourism, agriculture, fisheries, and maritime trade. Despite their economic and ecological value, the 2021 country and county climate risk profiles highlight that these resources are increasingly under threat from climate change impacts such as drought, flooding, sea level rise, ocean acidification, rising temperatures, and forest fires. The consequences of these risks exacerbate underlying socio-economic vulnerabilities, with women, youth, and Persons Abled Differently (PAD) experiencing disproportionate impacts due to structural inequalities. Loss of income, displacement, reduced productivity, and damage to infrastructure and markets are already affecting community well-being and resilience in the region.

In response, the COSME project aims to enhance women-led, climate-resilient livelihoods through ecosystem-based approaches. Hence, to support project implementation requires the development of Gender Responsive Adaptation and Preparedness Plans (GRAPPs) that integrate climate risk diagnostics with locally appropriate adaptation pathways. The Sokoke Ward Climate Change Adaptation Plan (WAP) is therefore prepared as part of the GRAPPs outputs. The ward which measures 102.7km^2, is located in Kilifi County at Kenya's Northern coast with a population of 14,238 (6,677 male & 7,561 female) (KNBS, 2019) faces a high vulnerability to climate change impacts. The ward's climate is a tropical coastal characterized by bimodal rainfall and warm temperatures year-round. Its economy and livelihoods depend heavily on climate-sensitive sectors such as rain-fed agriculture, livestock and forest resource use (Figure 1). However, recurring droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode development gains and degrade critical natural resources upon which local livelihoods are dependent, making adaptation a critical priority for the ward.

Recurring droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode hard-won development gains, making adaptation a critical priority for the ward. In response, Kenya's policy framework has evolved to support local climate action, from the National Climate Change Response Strategy and Action Plans to the Climate Change Act 2023, which emphasizes devolved climate governance and resilience building at county and community levels. This Ward Adaptation Plan aligns with national and county climate change policies by mainstreaming adaptation into local planning and empowering the community to implement priority resilience measures.

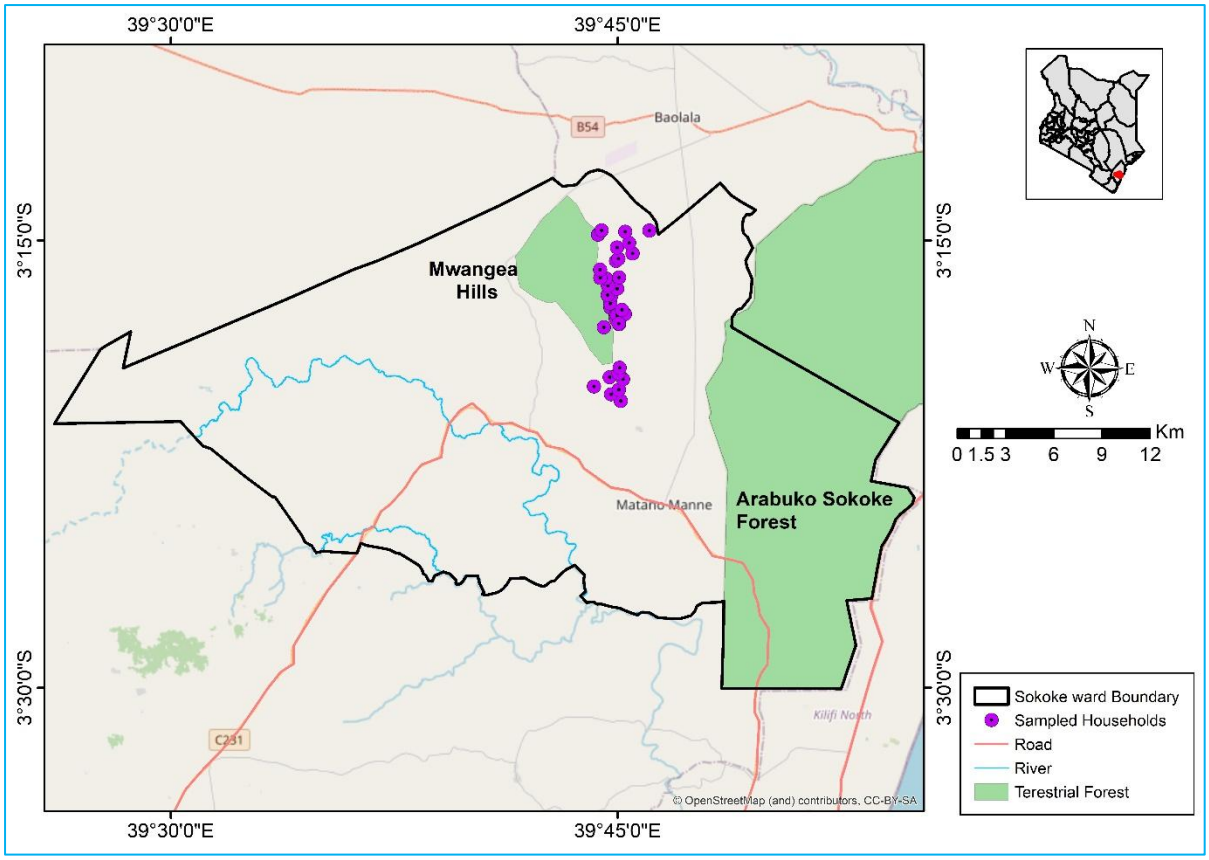

Figure 1: Location of Sokoke Ward and survey sites Source: PCVA, 2025)

1.2 Plan development approach

The Ward Adaptation Plan for Sokoke was developed in a participatory, gender-responsive and livelihoods-based vulnerability framework that conceptualized climate risk as an outcome of the interaction between exposure, sensitivity and adaptive capacity (Figure 2). Within this analytical framing, vulnerability was understood as the degree of climate-induced damage experienced by socio-ecological systems, while resilience denoted the system's ability to maintain or rapidly regain functionality following climatic shocks. This approach acknowledged that climate risks in Sokoke, characterized predominantly by recurrent droughts, seasonal flooding, strong winds, pests and diseases and human–wildlife conflicts arising from proximity to Sokoke forest and Mwangea hills, do not affect population groups uniformly. Women, men, youth, elderly persons and Persons Abled Differently (PADs) face differentiated risks due to disparities in livelihood assets, labor burdens, mobility, access to resources, intra-household power dynamics and decision-making autonomy.

The plan was therefore designed using a process that combined experiential community knowledge with quantitative datasets, spatial analysis, ecological diagnostics and climatic modelling. This integrated methodology enabled the plan to link climatic hazards to specific livelihood systems; rainfed agriculture, livestock keeping, small-scale trade and casual labour and to examine how these systems are shaped by gender norms, social vulnerabilities and institutional arrangements at ward level. The resulting insights provided a nuanced understanding of how climate extremes intersect with poverty, food insecurity, ecosystem degradation, weak infrastructure and limited institutional support, thereby shaping the context within which adaptation strategies must be developed.

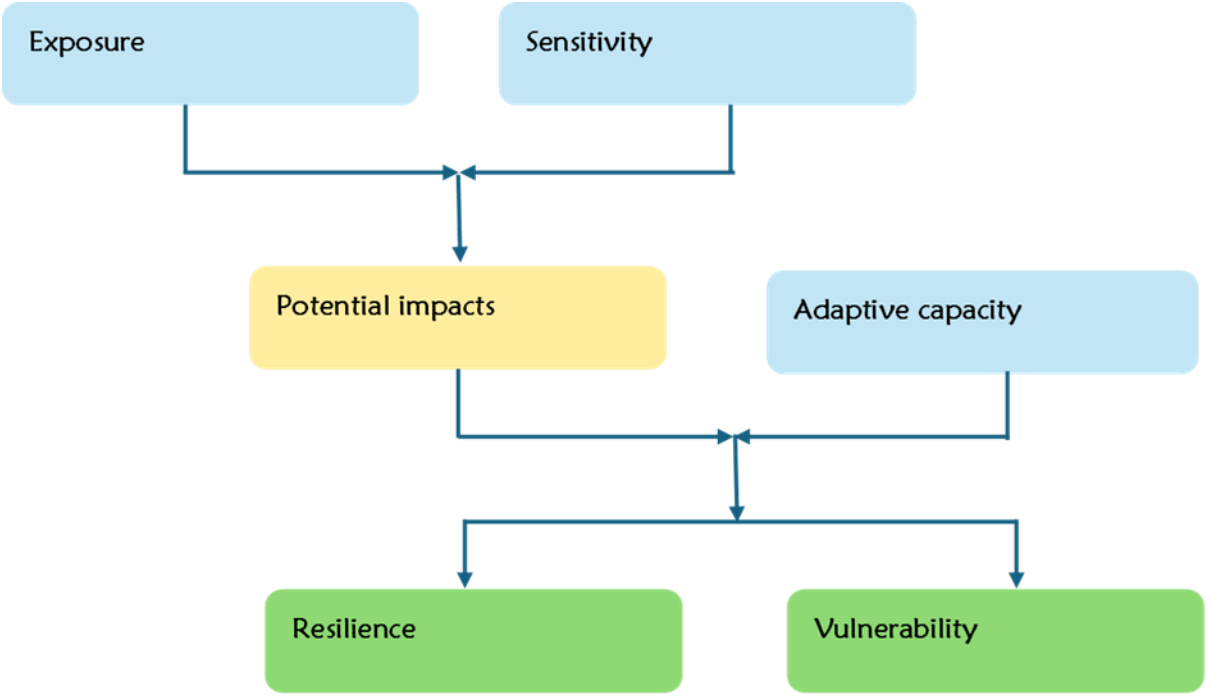

Figure 2: A model of factors that can be used to assess the vulnerability and resilience of systems (Source: Füssel and Klein, 2006, and Gauthier et al., 2014)

1.2.1 Inception Phase

The methodology was agreed upon through an inception phase that established a shared understanding among Plan International Kenya and the consultant on the objectives, scope and analytical expectations of the PCVA and WAP processes. An inception meeting held on 5th September 2025 brought together the COSME project teams to refine the methodological sequence, expected results, deliverable structure and timelines. During this phase, existing climate risk information for Kilifi County was reviewed, data gaps specific to Sokoke Ward were identified, and a sequenced work plan was developed outlining participatory diagnostics, household surveys, spatial and climate analysis, and subsequent synthesis and validation activities.

Following institutional alignment, data collection instruments were developed and technically refined to ensure that they were methodologically sound, gender-responsive and culturally appropriate. The instruments included ward-level FGD guides segregated for women, men and youth; participatory mapping and historical timeline guides; seasonal calendar and daily clock templates; household decision-making pile-sorting tools; Leaky Bucket diagrams; impact chain protocols; vulnerability matrix templates; institutional Venn diagram formats; and adaptation pathway worksheets. Each tool integrated prompts on gender norms, intersectional vulnerability, unpaid care workloads, labour shifts during climate stress and disability inclusion.

Stakeholder mapping during the inception phase identified relevant actors within Sokoke Ward, including the Ward Climate Change Planning Committee, chiefs and assistant chiefs, village elders, leaders of women and youth groups, PAD representatives and support networks, BMUs, WRUAs and community-based organizations. These actors facilitated community mobilization, venue preparation and dissemination of study information at ward level. The inception phase ensured that ward stakeholders were adequately oriented and positioned to meaningfully participate in the PCVA and adaptation planning process.

1.2.2 Participatory Climate Risk and Vulnerability Assessment (PCVA)

The Ward Adaptation Plan was informed by a comprehensive PCVA process implemented in four COSME target villages within Sokoke Ward i.e., Dzikudze, Midugani, Mwele and Ndungumnani. The assessment followed a structured six-step participatory sequence designed to diagnose climate vulnerabilities, identify underlying drivers of sensitivity, assess existing coping and adaptation strategies, and translate diagnostic findings into actionable adaptation options suitable for inclusion in the WAP. The process applied the full suite of ten PCVA tools, each deployed to generate gender- and age-disaggregated evidence capable of capturing complex interactions between climate hazards, livelihoods, natural resources and social systems.

1.2.2.1 Step 1: Scoping

During scoping, the geographical scope of the ward was delineated using GIS shapefiles provided by Plan International and Kilifi County Government. Within this boundary, the four COSME villages were purposively selected on the basis of ecological sensitivity, concentration of climate-sensitive livelihoods, recurrent exposure to droughts and floods, susceptibility to strong winds and documented human–wildlife conflict incidents, particularly elephant attacks reported around Dzikudze & Midugani villages were then developed to summarize demographic characteristics, livelihood systems, existing infrastructure, ecosystem conditions, climate-related impacts, past interventions and vulnerability hotspots. These profiles served as the baseline data against which PCVA tools were deployed.

1.2.2.2 Step 2: Establishment of Core Participatory Analysis Teams

A ward-level core participatory analysis team was established for Sokoke, drawing membership from Ward Climate Change Planning Committee (WCCPC) representatives, village elders, women leaders, youth leaders, PAD representatives and carers, NbS groups, GJJ groups, and other active CBOs and self-help groups. County technical officers from climate change, environment, forestry, agriculture and livestock were engaged as resource persons.

The core team underwent a structured orientation on the PCVA framework, the ten PCVA tools, gender- and age-responsive facilitation, child safeguarding and ethical data collection. Training sessions used a combination of plenary presentations, group work and simulations of each PCVA tool, conducted primarily in Kiswahili to ensure comprehension. By the end of the training, the ward team had the capacity to co-facilitate hazard and resource mapping, historical timelines, seasonal calendars, daily clocks, household decision making and pile-sorting exercises, Leaky Bucket analyses, impact chain mapping, vulnerability matrix scoring, Venn diagrams and adaptation pathway development.

1.2.2.3 Step 3: Situation Analysis of Climate Vulnerability

The situation analysis integrated secondary datasets, primary field evidence and participatory analytics to generate a comprehensive picture of exposure, sensitivity and adaptive capacity across Sokoke Ward. This phase synthesised literature review, ward-level gender- and age-disaggregated Focus Group Discussions (FGDs), transect walks across the five COSME villages, and the structured application of PCVA Tools 1–7. Through triangulation, these methods produced a detailed, spatially nuanced and socially differentiated understanding of how climatic hazards interact with biophysical systems, livelihood structures and gendered social dynamics.

1.2.2.3.1 Literature Review

The literature review served as the technical backbone of the situation analysis, drawing from global, regional, national, county and historical ward-level evidence to contextualize observed vulnerabilities. At the global scale, the review considered findings from IPCC AR6 Working Group I and II, particularly projections relevant to East African coastal systems, including

accelerated warming rates, sea surface temperature rise, intensification of short-duration extreme rainfall, increased occurrence of coastal storms, saline intrusion risks and anticipated sea-level rise along the Western Indian Ocean. These findings were aligned with global adaptation and risk governance frameworks such as the Sendai Framework for Disaster Risk Reduction 2015–2030, the Paris Agreement adaptation cycle, and SDGs 2, 5, 6, 13 and 14, which collectively frame resilience building in coastal and marine socio-ecological systems.

National climate policy and regulatory frameworks were examined comprehensively, including the Climate Change Act (2023), National Adaptation Plan (NAP 2015–2030), Kenya's Updated NDC (2023), the National Framework for Climate Services (NFCS), Kenya Climate-Smart Agriculture Strategy, National Disaster Management Policy, National Drought Early Warning System, and EMCA (2015) Cap 387 of the Laws of Kenya and sectoral regulations. These documents provided the lens for interpreting institutional mandates, priority hazards and adaptation responsibilities that cascade to county and ward levels.

At county scale, the review consolidated information from the Kilifi County Integrated Development Plan (CIDP III, 2023–2027), the County Climate Change Fund (CCCF) Regulations, the County Climate Change Action Plan (CCCAP 2023–2027), the County Spatial Plan, and sector strategies in agriculture, livestock, fisheries, water and coastal management. These county frameworks were triangulated with socio-economic and food security trends reported by the National Drought Management Authority (NDMA), historical rainfall data from the Kenya Meteorological Department (KMD), marine ecosystem studies from KMFRI, forestry and mangrove assessments by the Kenya Forest Service (KFS), fishery catch records from the State Department for Fisheries, and demographic data from the Kenya Population and Housing Census (2019).

The review also incorporated previous PCVAs conducted under COSME and other coastal adaptation programmes, which provided comparative vulnerability patterns, gender-responsive insights, and past lessons on nature-based solutions, community-based early warning systems, and local adaptation practices. This multilevel synthesis created a scientifically grounded baseline against which primary data from Sokoke could be interpreted.

1.2.2.3.2 Focus Group Discussions (FGDs)

Ward-level FGDs were conducted with segregated groups of women, men and youth from the four COSME villages; Dzikudze, Midugani, Mwele and Ndungumnani, to elicit gendered and generational insights into climate trends, livelihood impacts and socio-ecological vulnerability (Figure 3). The discussions explored perceived changes in rainfall reliability, including delayed onset and inconsistent cessation of rains, prolonged intra-seasonal dry spells, unprecedented periods of elevated temperatures, shifts in wind patterns and increased intensity of episodic storms. Participants described the material impacts of these climatic variations on agricultural productivity, livestock mortality and food security.

The FGDs also highlighted the deeply gendered nature of climate vulnerability. Women described increased walking distances and time burdens for water and fuel collection during droughts; heightened caregiving demands during disease outbreaks; increased exposure to GBV during water scarcity periods; and disruptions in women's informal trading activities. Youth reported loss of wage labour opportunities during climate shocks, stress-related migration, and lack of access to climate-resilient livelihood assets. Men described reduced crops and livestock yields and increased anxiety over household provisioning responsibilities. In addition, participants highlighted PAD-specific vulnerabilities, including mobility limitations during floods and exclusion from labour-intensive coping strategies.

These discussions produced climate calendars, gender-disaggregated vulnerability matrices and hazard trend narratives that deepened the temporal and social dimensionality of the vulnerability assessment.

Figure 3: Focus Group Discussions in Sokoke Ward (Source: Field Surveys,2025)

1.2.2.3.3 Transect Walks

Transect walks undertaken across the four villages provided on-ground spatial verification of vulnerabilities identified in FGDs and documented in literature. These walks traversed rainfed agricultural plots, Mwangea hills, Kaya kadzitsoni, boreholes and shallow wells, public institutions (schools), access roads and market areas.

Structured observation sheets and GPS-enabled devices were used to capture geospatial attributes of key vulnerability hotspots. Evidence recorded included deterioration of feeder roads due to flooding, heavily eroded gullies formed by runoff from bare catchments, degraded Mwangea hills exhibiting canopy thinning or sediment stress, and infrastructure damage such as submerged pathways.

These observations enhanced the physical realism of the vulnerability analysis and provided geo-referenced evidence that later strengthened hazard–resource maps and the identification of priority zones for ecological restoration and climate-resilient infrastructure.

1.2.2.3.4 Tool 1: Hazard and Resource Mapping

The hazard and resource mapping tool enabled community members to visually represent their environmental and socio-economic landscape (Figure 4). Through participatory sketching, communities identified settlement clusters, farmlands, Mwangea hills, kaya kadzitsoni, markets, schools, water points and religious centres. Hazard features; such as recurrent pest & diseases outbreaks, strong-wind corridors and elephant-risk stretches were marked and discussed. This

mapping exercise generated spatially explicit knowledge that later informed adaptation targeting and zoning.

Figure 4: Hazard and Resource Mapping in Dzikudze Village, Sokoke Ward (Source: PCVA, 2025

1.2.2.3.5 Tool 2: Historical Timelines

Historical timelines reconstructed multi-decadal climatic and socio-economic events including major droughts, destructive floods, pest invasions (armyworms and whiteflies), severe farm losses from temperature anomalies and episodes of heightened human–wildlife conflict. Each event was discussed in terms of magnitude, social and economic impact, community responses and institutional support. Timelines revealed a pattern of intensifying climatic shocks, shortened recovery cycles and cumulative stress on natural resource systems.

1.2.2.3.6 Tool 3: Seasonal Calendars

Seasonal calendars detailed intra-annual cycles of rainfall, temperature, wind regimes, agricultural operations, disease outbreaks and social events. Comparisons between historical and current seasonal behaviour revealed declining predictability of rainfall, increased occurrence of off-season rainfall and persistent dry spells. Calendars also captured periods of heightened GBV, school absenteeism, food scarcity and increased care burdens.

1.2.2.3.7 Tool 4: Daily Clock

The daily clock exercise revealed gendered labour allocation and how climate stress reconfigures time use. Under climate-normal conditions, tasks such as agricultural operations, childcare, water collection and marketing followed relatively stable patterns. Under climate stress, women and girls reported significant increases in unpaid care work and water collection time, men noted prolonged periods of income loss due to weather-inhibited farming, and youth described increased domestic workload substitution and reduced learning hours. These insights demonstrated how climate shocks deepen time-poverty and reduce adaptive capacity.

1.2.2.3.8 Tool 5: Household Decision-Making Pile Sorting

Pile sorting exercises clarified intra-household power/authority structures under normal and climate-stress scenarios. Under normal conditions, decision-making on crop choices, livestock sales and minor household investments tended to be joint or gender-specific depending on cultural norms. Under climate stress, however, the decision space often shifted toward male dominance, particularly in decisions involving financial risk, asset disposal or climate-response investment, which in turn constrained women's adaptive agency and reinforced gender-based vulnerabilities.

1.2.2.3.9 Tool 6: Leaky Bucket

The Leaky Bucket tool illustrated how climate shocks disrupt household financial stability. Communities mapped income streams such as agriculture, petty trade, remittances and casual labour, alongside expenditure flows including food purchases, health costs, water, school fees, debt servicing and transport. Climate shocks were shown to reduce inflows drastically while simultaneously raising outflows through rising food prices, repair costs and medical expenses. This widening income–expenditure gap demonstrated the structural economic vulnerability of households.

1.2.2.3.10 Tool 7: Impact Chains

Impact chain analysis depicted the cascading consequences (direct and indirect impacts) of priority hazards. For instance, prolonged drought triggered crop losses, water scarcity, livestock mortality and reduced milk intake, which in turn contributed to household food insecurity, malnutrition risk among children, increased labour migration by men, rising care burdens for women, heightened GBV risk and reduced school attendance. Similarly, pest and diseases disrupted farming, leading to economic losses, debt accumulation and psychological stress. This tool exposed multi-dimensional vulnerability pathways and supported the design of integrated adaptation strategies.

1.2.2.4 Step 4: Analysis of Causes of Vulnerability (Tool 8 – Vulnerability Matrix)

The analysis of underlying causes of vulnerability was undertaken through the application of the vulnerability matrix (Tool 8), which provided a structured means of examining how different climatic hazards intersect with livelihood assets and socio-ecological systems in Sokoke Ward. Hazards such as drought, floods, strong winds, human–wildlife conflict and agricultural pests and diseases were arranged against critical assets including rain-fed farms, Mwangea hills, livestock, domestic water sources, housing structures, schools, health facilities, feeder roads and market infrastructure. Community members assigned severity ratings for each hazard–asset interaction, supported by qualitative reasoning and lived examples.

The cumulative scoring process produced a ranking that demonstrated, for example, the acute vulnerability of rain-fed agriculture to prolonged drought and unpredictable rainfall; and the differentiated impacts of floods on schools and feeder roads which disrupt mobility, market access and educational continuity. This step bridged descriptive hazard analysis with deeper socio-economic and institutional determinants of vulnerability, forming the foundation for identifying priority adaptation pathways.

1.2.2.5 Step 5: Analysis of Community Action (Tool 9 – Venn Diagrams)

Analysis of existing coping and adaptation practices revealed a range of coping strategies and externally supported interventions. The Venn diagram exercise (Tool 9) enabled communities to map the institutional ecosystem influencing adaptation in the ward. Institutions such as county line departments, NGOs, community-based organisations, religious institutions, savings and credit groups, COSME project structures, and private sector actors were placed on diagrams according to their perceived influence and accessibility. The spatial arrangement revealed a

significant presence of institutions but varying degrees of functional support. While some organisations maintained close, supportive relationships, such as community groups collaborating with NbS, others operated at a distance, limiting their impact. This analysis also identified critical gaps, including weak linkages to financial services, limited climate information dissemination, and inadequate institutional frameworks for systematic PAD inclusion. These findings provided essential insights for designing governance and institutional strengthening elements within the Ward Adaptation Plan.

1.2.2.6 Step 6: Drawing Action from Analysis (Tool 10 – Adaptation Pathways)

The final step in the participatory diagnostic process involved translating identified vulnerabilities, exposure pathways and institutional gaps into a coherent adaptation strategy through the Adaptation Pathways tool (Tool 10). Communities revisited impact chains and vulnerability dynamics to identify feasible response strategies across short-, medium- and long-term horizons. The exercise involved structured deliberations that considered gendered labour implications, the availability of financial and natural resources, household and community capacities, and alignment with ongoing COSME and county initiatives.

Resulting adaptation pathways articulated sequenced actions for key socio-ecological systems; agriculture, water resources, terrestrial forests, community infrastructure, early warning systems and social protection mechanisms. This included immediate resilience-building measures such as water harvesting, and household-level livelihood diversification; medium-term actions such as Provision of farm inputs (certified seed varieties for drought tolerant crops, organic and inorganic fertilizers, pesticides) and strengthening local governance structures; and long-term interventions such as rehabilitation of Mwangea hills, institutionalization of community-based climate monitoring systems, and improvement of rural infrastructure. These pathways formed the backbone of the Sokoke Ward Adaptation Plan, linking diagnostic insights to strategic action.

1.2.3 Household Survey

A structured household survey was administered to 295 households in the study area to provide quantitative evidence complementing qualitative findings. Sampling was designed to include households across diverse livelihood portfolios and different vulnerability categories, including female-headed households, large households with high dependency ratios, households with PAD members, and those primarily reliant on climate-sensitive livelihoods such as rain-fed agriculture. Primary respondents were predominantly women (approximately 65 %), ensuring robust capture of intra-household dynamics, care burdens, access to services and lived experiences of climate stress.

The survey collected data on demographic structure, education, disability status, land tenure and resource access, livelihood contributions, income diversification, exposure to recent climatic shocks, losses and damages, coping strategies, food security indicators, institutional linkages and participation in social protection and savings groups. Data were collected digitally using KoboToolbox to ensure consistency and reduce transcription errors. These data provided a statistically grounded basis for validating PCVA findings and understanding patterns of vulnerability at household level across gender, age and livelihood categories.

1.2.4 Forest Health Assessment

To assess the condition of key coastal and terrestrial ecosystems supporting nature-based adaptation in Sokoke, a remote sensing analysis was conducted using Sentinel-2 imagery at 10-metre spatial resolution. Imagery underwent standard pre-processing steps including radiometric correction and cloud masking before the Normalized Difference Vegetation Index (NDVI) was computed using the formula.

$$NDVI = \frac{NIR - RED}{NIR + RED} \qquad \text{Equation 1}$$

Where RED is visible red reflectance, and NIR is near infrared reflectance.

NDVI values were categorised into low, moderate and high vegetation vigor classes. Low NDVI signatures corresponded to degraded or stressed vegetation, often reflecting factors such as saltwater intrusion, wood extraction pressure, sedimentation and hydrological alteration. High NDVI zones served as ecological benchmarks, indicating healthier vegetation stands. These spatial patterns were subsequently compared with PCVA hazard–resource maps and field observations to identify priority sites for ecological restoration and sustainable forest management within the ward.

1.2.5 Current and Future Climate Projections

Climate projections relevant to Sokoke were developed using downscaled county-level analyses supplemented by remote sensing and historical observational datasets. Daily rainfall Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS), Actual Evapotranspiration (AET), Potential Evapotranspiration (PET), (from ERA5-Ag) and temperature variables (minimum, mean, maximum and Land Surface Temperature (LST) from Landsat and ERA5-Ag) were extracted through Google Earth Engine after spatial boundaries had been validated in QGIS. The resulting datasets were aggregated into annual totals or means and subsequently analysed using Python (Colab) employing pandas, numpy, scikit-learn and matplotlib.

Linear regression models were fitted to each climatic variable to estimate historical trends and project future changes to 2035 and 2045. Temperature projections incorporated Intergovernmental Panel on Climate Change (IPCC) AR6-aligned warming increments for East Africa's coastal belt, applying approximately +0.02°C per year for the moderate-emission scenario and +0.035°C per year for high-emission trajectories. Rainfall projections followed empirically derived trends, showing increasing variability, changes in seasonal distribution and heightened likelihood of intense episodic rainfall events. These climatic insights were interpreted in relation to Sokoke's hazard profile, reinforcing the need for enhanced water security, climate-resilient agricultural practices, stronger infrastructure planning and expanded early warning systems.

1.2.6 Data Analysis and Reporting

Quantitative data from the household surveys were cleaned, coded and analyzed using Microsoft Excel and Python. Analysis focused on frequencies, proportions, cross-tabulations and simple correlations across gender, age group, livelihood type and vulnerability status. These results provided empirical confirmation of patterns emerging from qualitative insights.

Qualitative data from FGDs, PCVA exercises and transect walks were transcribed and thematically coded to organise evidence under key domains such as hazard exposure, livelihood impacts, gender dynamics, coping strategies, institutional barriers and opportunities. Outputs from PCVA tools; maps, timelines, calendars, clocks, matrices and impact chains, were digitized and integrated with NDVI layers and climate trend visualizations to produce synthesized analytical outputs. Triangulation across all sources ensured internal consistency and improved the robustness of the final interpretation. Draft findings were compiled for review by Plan International Kenya in line with expected deliverables.

1.2.7 Validation

The draft findings were presented in county-level validation workshops held in Kilifi and Kwale on 20th and 21st November 2025 respectively. These sessions provided a platform for county technical departments, CCPC/WCCPC members, civil society, COSME partners and community representatives to review the PCVA results and refine the draft adaptation strategies. At ward level, stakeholders from Sokoke reviewed hazard rankings, adaptation priorities and institutional roles, and provided further refinement on feasibility, gender inclusion, sequencing and resource implications. Inputs from these processes were incorporated into the final Ward Adaptation Plan to ensure scientific accuracy, community legitimacy and institutional alignment.

2 Chapter 2: Coordination of Adaptation Actions

2.1 Introduction

Effective coordination is essential in transforming climate adaptation goals into tangible resilience outcomes. In Sokoke Ward, climate risks affect various sectors such as agriculture, water resources, health, ecosystems, and infrastructure. The impact of these risks differs among social groups, with women, youth, and other vulnerable populations often facing the greatest hardships. Implementing the Ward Adaptation Plan therefore demands a governance framework that connects community priorities with county planning structures and national climate policies, while ensuring that local actors have access to finance, technical expertise, and reliable climate information. This chapter outlines that coordination framework and its linkage to Kenya's National Adaptation Plan (NAP) 2015–2030. The coordination mechanism emphasizes both horizontal collaborations, bringing together village-level institutions, community-based organizations, private entities, and technical officers within the ward and vertical integration, ensuring that ward priorities are embedded in county climate change plans and budgets, and aligned with the NAP's strategic objectives. These include climate-smart agriculture, sustainable water resource management, ecosystem restoration, disaster preparedness and early warning systems, and health resilience. The outcome is a harmonized structure where institutional roles are clearly defined, duplication of efforts is minimized, and locally led, gender-responsive adaptation initiatives are effectively implemented, monitored, and sustained over time.

2.2 Current Institutional Arrangements

Coordination of climate adaptation in Sokoke Ward operates through three interconnected levels, ensuring alignment between national guidance, county strategies, and local implementation. At the national level, the Ward's adaptation actions are anchored in the Climate Change Act 2023, which provides the legal framework for devolved climate governance, and guided by the National Adaptation Plan (NAP), 2015–2030 and the National Climate Change Action Plan (NCCAP) 2023–2027. These instruments define national priorities, provide technical guidance, and establish standards for adaptation interventions. Specialized agencies provide technical support : the Kenya Meteorological Department delivers weather forecasts and impact-based warnings crucial for agriculture, fisheries, and disaster preparedness; the National Drought Management Authority coordinates early warning systems and triggers for social protection during drought events; the Kenya Forest Service and Kenya Wildlife Service provide support for forest restoration, biodiversity management, and mitigation of human-wildlife conflicts; while the National Environment Management Authority (NEMA) ensures that adaptation investments comply with environmental safeguards. These agencies engage Sokoke Ward mainly through county structures, with direct involvement during extreme events like droughts.

At the county level, the County Government of Kilifi translates national directives into actionable local interventions. The Kilifi County Climate Change Policy, 2023, the Kilifi County Climate Change Action Plan, 2023–2027, The Kilifi County Climate Change Act, 2023 provide the strategic and operational framework for coordinating adaptation efforts across the county. These instruments embed climate considerations into the County Integrated Development Plan (CIDP 2023–2027) and sectoral plans, including the Departments of Agriculture, Livestock Development & Blue Economy; Health and Sanitation Services; Environment, Forestry and Natural Resources; Lands, Physical Planning; Housing and Urban Development; and Public Service, Special Programmes and Disaster Management; ensuring resilience is integrated across all key sectors.County coordination platforms, such as the County Climate Steering Committee and the Disaster Management Forum, oversee multi-hazard preparedness and enable anticipatory planning based on seasonal climate forecasts. Partnerships with non-governmental

organizations and development partners are convened through county-led forums so that external programming in Sokoke is harmonized with public investments and avoids duplication.

At the ward and community level, implementation is driven by locally led structures that link planning, decision-making, and on-the-ground action. The Ward Administrator acts as the administrative liaison to county directorates, while the Ward Climate Change Planning Committee (WCCPC) functions as the main coordination body for Sokoke Ward. The WCCPC includes local administration, technical extension officers, Water User Associations, Community-Based Organizations, and representatives of women, youth, PADs and respected elders. The committee guides investment prioritization, monitors progress, and facilitates community engagement in adaptation actions. Village-level structures, such as water user committees, disaster volunteer teams, and environmental stewardship groups, execute day-to-day interventions, including maintaining boreholes and managing forest restoration projects. By formalizing these linkages, the WCCPC strengthens communication, ensures accountability, and promotes inclusive decision-making, enabling Sokoke Ward to implement adaptation measures effectively while fostering community ownership.

2.3 Relevant Policies

The policy environment guiding coordination is robust and multi-scalar. The Climate Change Act (2023) mandates climate integration across public planning, establishes coordination bodies, and provides the legal basis for devolved climate finance. The NAP 2015–2030 defines national priorities for adaptation and sets out sectoral action areas that are directly germane to Sokoke: strengthening integrated water resources management and water supply resilience; scaling climate-smart agriculture and resilient livestock systems; safeguarding health through surveillance and risk communication; climate-proofing infrastructure and human settlements; protecting and restoring forests, and institutionalizing risk-informed development through early warning and disaster risk reduction. The NCCAP 2023–2027 operationalizes these priorities into programs and financing pathways, notably the Financing Locally Led Climate Action (FLLoCA) program that channels performance-based climate grants to county and ward levels.

Sub-nationally, the Kilifi County Climate Change Act (2021) and the County Climate Change Action Plan (2023–2027) provide an enabling framework for ward proposals to access the County Climate Change Fund and align with county targets on water security, resilient agriculture, coastal and riparian restoration, and climate-proofed infrastructure. Sector-specific policies reinforce these mandates: the Water Act and National Water Policy embed community-based water governance and integrated water resources management; the Agricultural Sector Transformation and Growth Strategy promotes diversified, climate-smart production and value addition; the National Disaster Risk Management policy architecture advances anticipatory action and local preparedness; the Wildlife Conservation and Management Act provides mechanisms for human–wildlife conflict mitigation and compensation; and national gender and social inclusion policies require equitable participation and benefit sharing. Together, these instruments establish the principles, standards, financing mechanisms, and safeguards that the Sokoke coordination system adopts, and thereby ensuring consistency with the NAP while tailoring execution to place-based realities.

2.4 Proposed Coordination Arrangements

The proposed coordination arrangements build on existing structures while establishing clear roles, communication channels, and operational routines to align the Ward's adaptation efforts with NAP principles of coordination, local empowerment, and continuous learning. At the ward level, the Ward Climate Change Planning Committee (WCCPC) will serve as the central coordination body, chaired by the Ward Administrator. The committee will oversee the planning, implementation, and monitoring of all adaptation initiatives in Sokoke Ward. A

County Climate Change Officer will act as a formal link between the WCCPC and county structures, providing technical guidance, facilitating access to the County Climate Change Fund (CCCF), and ensuring alignment with broader county priorities. Progress will be tracked through quarterly reports, documenting outputs, challenges, gender-disaggregated participation, and adherence to environmental and social safeguards, thereby promoting accountability and a performance-driven approach.

At the village level, Village Adaptation Focal Points (VAFPs) in Ndungumnani, Mwele, Midugani, and Dzikudze will mobilize communities, report climate hazards, and coordinate localized adaptation measures. Implementation will be organized through task-specific teams, each combining technical expertise and community oversight. Water management teams will include Water Resource User Association members, village leaders, county officers, and partner representatives, while ecosystem restoration teams will bring together KFS staff, Community Forest Association (CFA) members, women's groups, and local conservation champions. Teams will maintain work logs, track resources, and provide monthly updates to the WCCPC.

A biannual ward-level stakeholder forum will bring together county departments, national agencies, NGOs, CBOs, CFAs, research institutions, faith-based organizations, and private-sector partners. The forum will map ongoing activities, set technical standards, coordinate complementary roles across villages, and identify implementation challenges. Between forums, the WCCPC will hold monthly coordination meetings to manage near-term tasks, including community mobilization, procurement sequencing, safeguards monitoring, and integration of seasonal forecasts and early warning information.

Adaptation coordination will be integrated with disaster risk management, ensuring seamless response to droughts, human-wildlife conflicts, pests and diseases and other hazards. The WCCPC will closely work with the Ward Disaster Management Committee to align long-term resilience building with acute event response. Protocols will guide how Kenya Meteorological Department advisories and NDMA triggers are interpreted and disseminated, and how ward teams coordinate with county emergency units for water provision, evacuation, or health campaigns.

Finally, flexibility and learning are embedded in the system. An annual review, co-convened by the WCCPC and the County Climate Change Unit, will assess progress against NAP-aligned indicators, document lessons learned, and adjust institutional arrangements or procedures. This ensures Sokoke Ward's adaptation coordination remains adaptive, inclusive, and responsive to both evolving policy guidance and local community needs.

2.5 Implementation Roles

Successful implementation of Sokoke Ward's Adaptation Plan requires clear delineation of responsibilities among all stakeholders to ensure that adaptation measures are executed efficiently, inclusively, and in alignment with Kenya's National Adaptation Plan (2015–2030). Table 1 below outlines the mandates, decision-making authority, and accountability mechanisms of the key institutions and actors engaged in rolling out adaptation interventions in Sokoke Ward. The WAP explicitly mainstreams Gender Equality and Inclusion (GEI) across all stages of planning and implementation and establishes strong coordination linkages with health, gender, and protection actors to guarantee continuity of essential Sexual and Reproductive Health and Rights (SRHR) and GBV prevention and response services during climate shocks

Table 1: Institutional Roles in Implementation of Adaptation Actions

Institution / Actor	Key Roles and Responsibilities
Ward Climate Change Planning Committee (WCCPC)	Lead planning, coordination, and prioritization of ward-level adaptation actions; mobilize community participation; ensure gender and social inclusion in project design and implementation.
Kilifi County Government Departments	Provide technical oversight, sectoral expertise, and finance (Departments of Water, Agriculture, Livestock, Fisheries, Roads, Health, Environment and Forestry); integrate ward priorities into departmental workplans and budgets.
National Drought Management Authority (NDMA)	Lead drought monitoring, early warning, and response; integrate Sokoke Ward into national drought early warning systems; coordinate emergency relief when thresholds are reached.
Kenya Meteorological Department (KMD)	Provide localized weather and climate forecasts, early warnings, and advisories to support decision-making in agriculture, fisheries, and disaster preparedness.
Kenya Forest Service (KFS)	Provide technical guidance on reforestation, afforestation; support tree nursery establishment and monitoring.
Kenya Wildlife Service (KWS)	Manage and mitigate human–wildlife conflicts; support establishment of water points for wildlife; train community wildlife scouts.
National Environment Management Authority (NEMA)	Review and approve Environmental Impact Assessments (EIAs) for adaptation projects; ensure adherence to environmental safeguards.
Development Partners (UNDP, GCF, World Bank, EU, FAO, IFAD)	Provide technical assistance, capacity building, and climate finance; support mainstreaming of climate resilience in local development and infrastructure projects.
Non-Government Organizations (Plan International, World Vision Kenya, WWF-Kenya, YISA)	Implement community-based adaptation projects; promote gender-responsive resilience initiatives; facilitate training on nature-based solutions (NbS) and livelihood diversification.
Community-Based Organizations (CBOs) and Self-Help Groups	Implement localized adaptation actions such as mangrove restoration, water harvesting, and livelihood diversification; mobilize local labor and indigenous knowledge; ensure equitable participation in project benefits.
Private Sector and Cooperatives	Promote investment in climate-smart agriculture, renewable energy, and insurance products; integrate sustainability in operations; support CSR initiatives for community resilience.
Academic and Research Institutions (KMFRI, KEMFRI, Universities)	Generate and disseminate climate and ecological data; support participatory research and innovation for adaptation; build capacity of local institutions on ecosystem monitoring and early warning systems.

3 Chapter 3: Situation analysis of climate vulnerability

3.1 Introduction

The Sokoke ward is increasingly experiencing multiple, interlinked, and climate-induced hazards whose intensity, duration, and frequency have grown over time. Findings from the Participatory Climate Vulnerability and Capacity Assessment (PCVA, 2025) indicate that the community is recurrently affected by droughts, Human–wildlife Conflicts (HWCs), environmental degradation, pests and diseases, floods and whirlwinds. These hazards have progressively disrupted ecosystem balance, altered natural resource functions, and reshaped traditional livelihood systems across the ward. The assessment employed a combination of community-based knowledge, historical trend analysis, and scientific projections to establish the frequency, intensity, and spatial extent of each hazard. Vulnerability was analyzed across the four sampled villages i.e., Ndungumnani, Mwele, Midugani, and Dzikudze, based on two key parameters: exposure to climatic stressors and adaptive capacity at household and institutional levels

The findings indicate that women, widows, the elderly, Persons Abled Differently (PADs) and low-income households are the most affected by climate-related impacts. Their heightened vulnerability is linked to their dependence on climate-sensitive natural resources, limited access to financial and productive assets, and the increased burden of unpaid care work such as caregiving and resource collection during periods of crisis. Spatial analysis further reveals that drought and human–wildlife conflicts are most pronounced in Dzikudze, Jila, Milole, Mlima wa Ndege, Ndugumnani, and Kwa Dadu, while pest infestations, crop diseases, and whirlwinds are prevalent across the entire ward. Flooding is mainly experienced in low-lying areas of Milore, Rare, Muriachakwe, Katangani, and Nyare.These spatial and demographic insights provide a critical foundation for designing targeted adaptation interventions that reduce vulnerability, minimize exposure, and strengthen the resilience of both communities and ecosystems in Sokoke Ward.

3.2 Current climate trends

Sokoke Ward lies within Kenya's coastal climatic zone, characterized by warm temperatures and two distinct rainfall seasons, the long rains from March to May and the short rains from October to December. The current climatic conditions in Sokoke Ward based on the baseline analysis under the SSP2–4.5 scenario, indicate a wet environment with annual rainfall averaging 1,136 mm, against a potential evapotranspiration (PET) of 1,622.1 mm and an actual evapotranspiration (AET) of 1,793.8 mm (Figure 3). The consistently higher PET values compared to rainfall demonstrate that atmospheric water demand exceeds precipitation input, predisposing the area to periodic moisture deficits and hydrological stress, especially during the dry months. Despite the relatively high rainfall totals, much of the moisture is rapidly lost through evaporation and plant transpiration, reducing effective water availability for crops, forests, and groundwater recharge.

Temperature analysis under the same scenario (Figure 4) shows a mean air temperature of 26.2°C and a land surface temperature (LST) of 35.2°C, indicating significant surface heating across terrestrial landscapes. The pronounced difference between air and surface temperatures reflects limited vegetation cover and increasing land exposure, which amplify heat absorption and retention. These warming conditions increase evapotranspiration, reduce soil moisture, and heighten heat stress on crops, livestock, and communities, underscoring the need for vegetation restoration and climate-smart land management in Sokoke Ward.

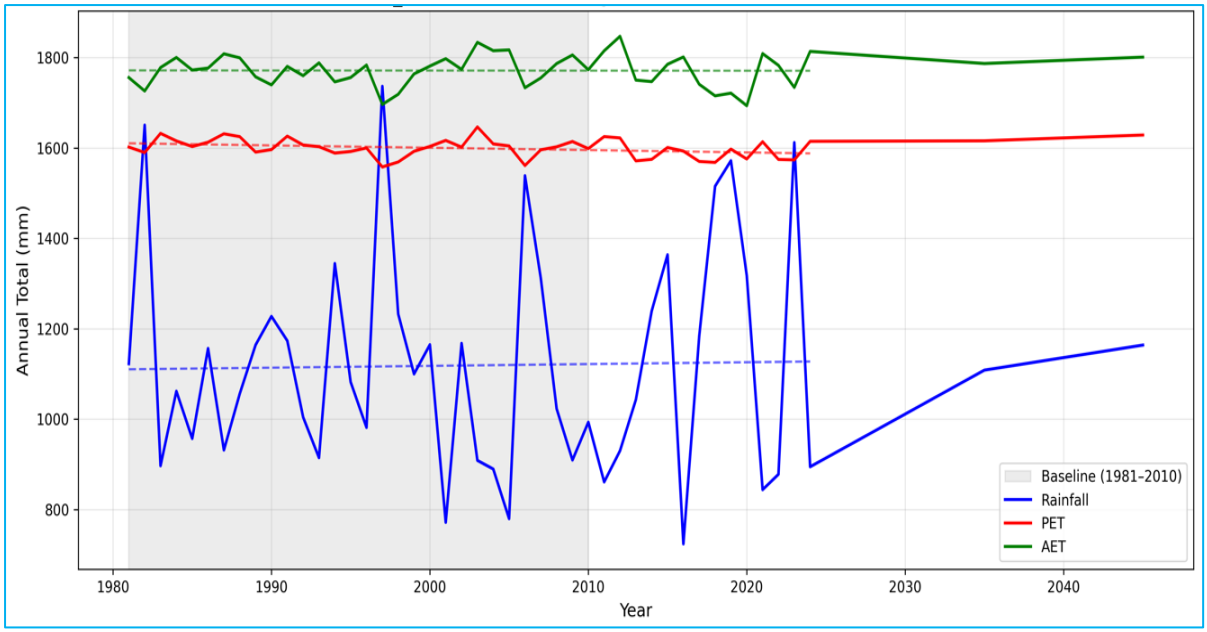

Figure 5: SSP2 4.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022)

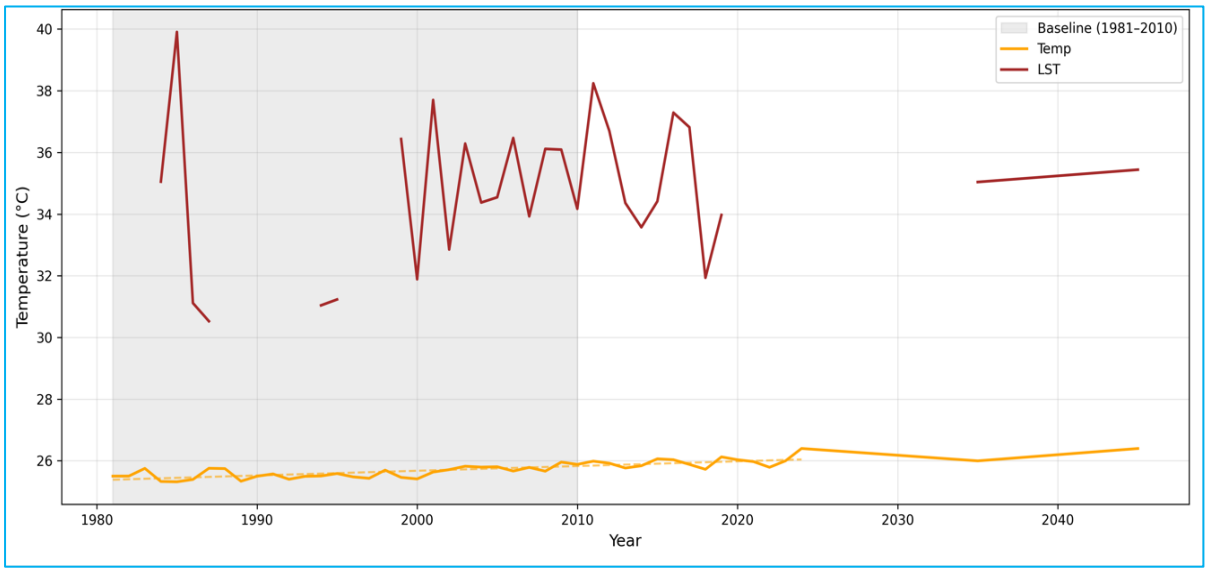

Figure 6: SSP2 4.5 Mean Temperature and Land Surface Temperature Projections. (Source: Chirps database and ERA5-Ag, 2022)

3.3 Future climate trends

Projected climate conditions under the high-emission scenario (SSP5–8.5) indicate that Sokoke Ward will experience slightly wetter but considerably warmer conditions relative to the baseline period. Rainfall is projected to increase from 1,141.5 mm to 1, 21.1 mm, PET from 1,631.7 mm to 1,650.9 mm, and AET from 1,804.4 mm to 1,825.7 mm (Figure 5). Despite higher rainfall, increased evapotranspiration may intensify soil moisture stress and heat impacts, highlighting the need for water management, afforestation, and climate-resilient agricultural practices.

Future temperature projections further emphasize the warming trajectory. The mean air temperature is expected to rise from 26.2°C to 26.6°C, while the LST increases significantly from 35.2°C to 35.6°C by 2045 (Figure 6). The strong surface heating effect reflects enhanced radiative forcing associated with greenhouse gas accumulation and local land degradation processes. Such temperature increases are expected to intensify evapotranspiration rates, accelerate crop water loss, and increase human and livestock heat stress.

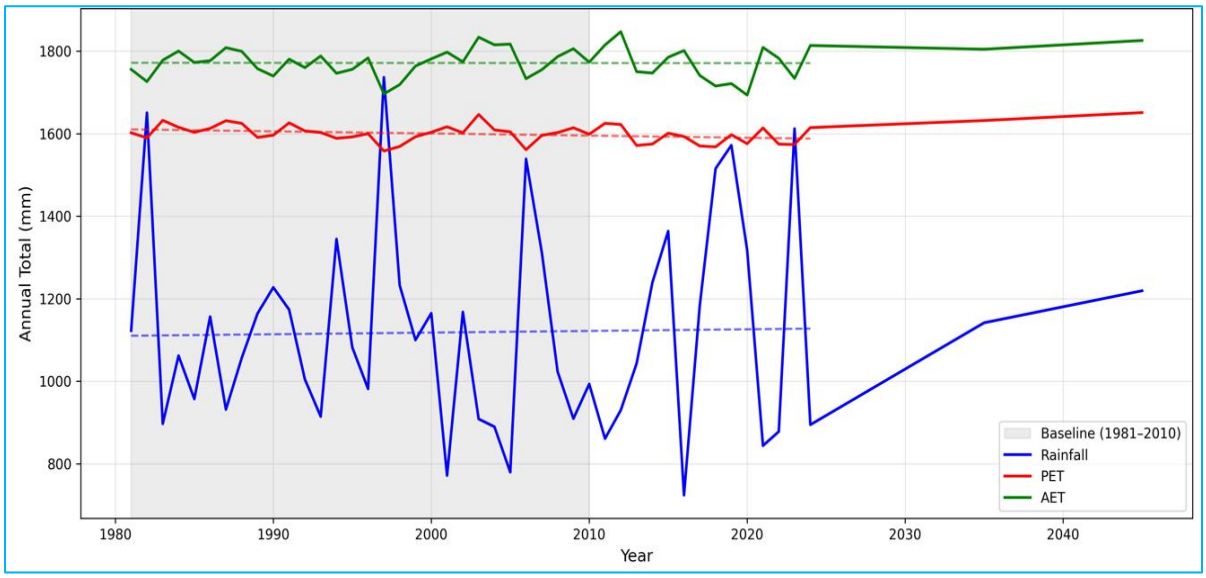

Figure 7: SSP5 8.5 Rainfall, AET and PET Projections. (Source: Source: Chirps database and ERA5-Ag, 2022)

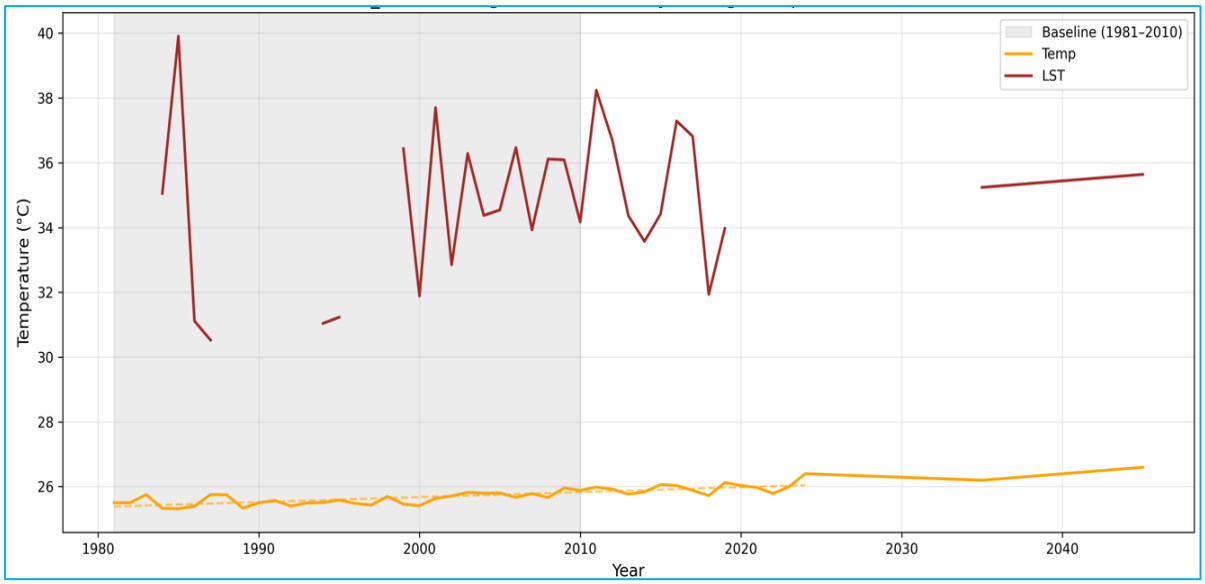

Figure 8 : SSP5 8.5 Mean Temperature and Land Surface Temperature Projections. (Source: Source: Chirps database and ERA5-Ag, 2022)

3.4 Climate hazards and vulnerability

3.4.1 Drought

Drought remains the most persistent and economically devastating climate hazard in Sokoke Ward, placing sustained pressure on both terrestrial and coastal ecosystems. Historical climate records from 1930 to 2025 indicate that major drought events occur approximately once every three (3) years, with notable severe episodes recorded in 1947, 1978, 1979, and 2013 (Figure 7). Over time, the frequency, duration, and intensity of these dry spells have increased, reflecting a marked decline in rainfall reliability and prolonged inter-seasonal dry periods. This trend is closely linked to delayed rainfall onset, early cessation of the long rains, and rising evapotranspiration rates, all of which have contributed to shorter growing seasons and recurrent crop failures.

Vulnerability levels vary considerably across the ward. Dzikudze (score 46) and Mwele (44) record the highest exposure, largely attributed to their reliance on rain-fed agriculture, proximity to degraded forest areas, and scarcity of reliable water sources. Ndugumnani (40) also faces significant vulnerability, whereas Midugani (20) experiences relatively lower impacts due to the availability of shallow wells and a more diversified livelihood base (Figure 8).

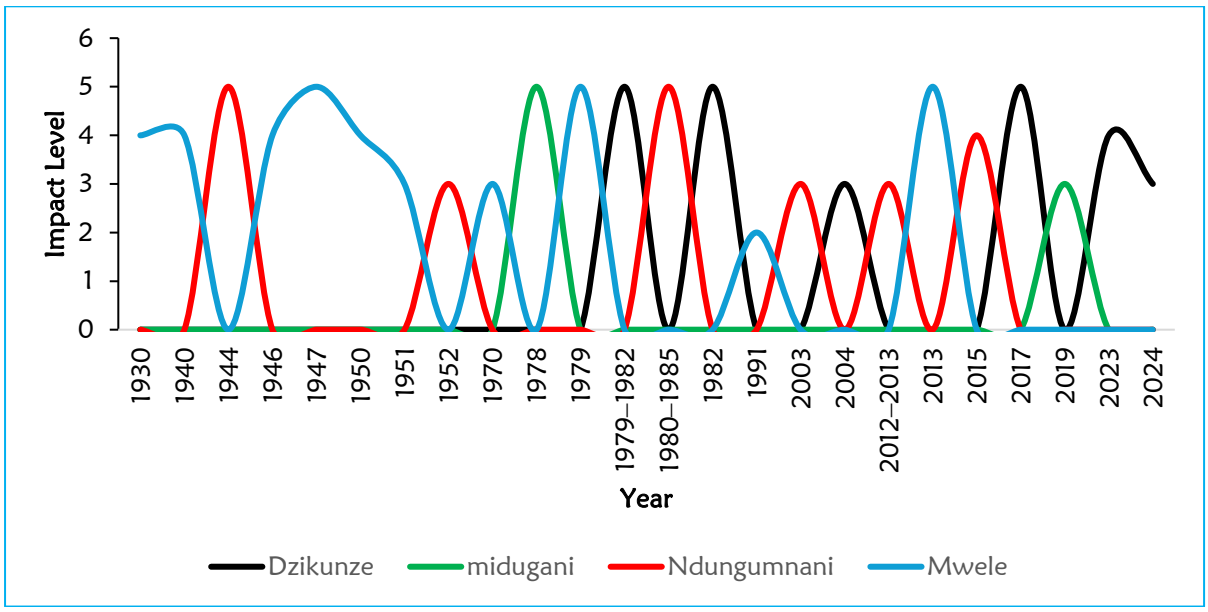

Figure 9: Drought prevalence historical trend (Source: PCVA, 2025)

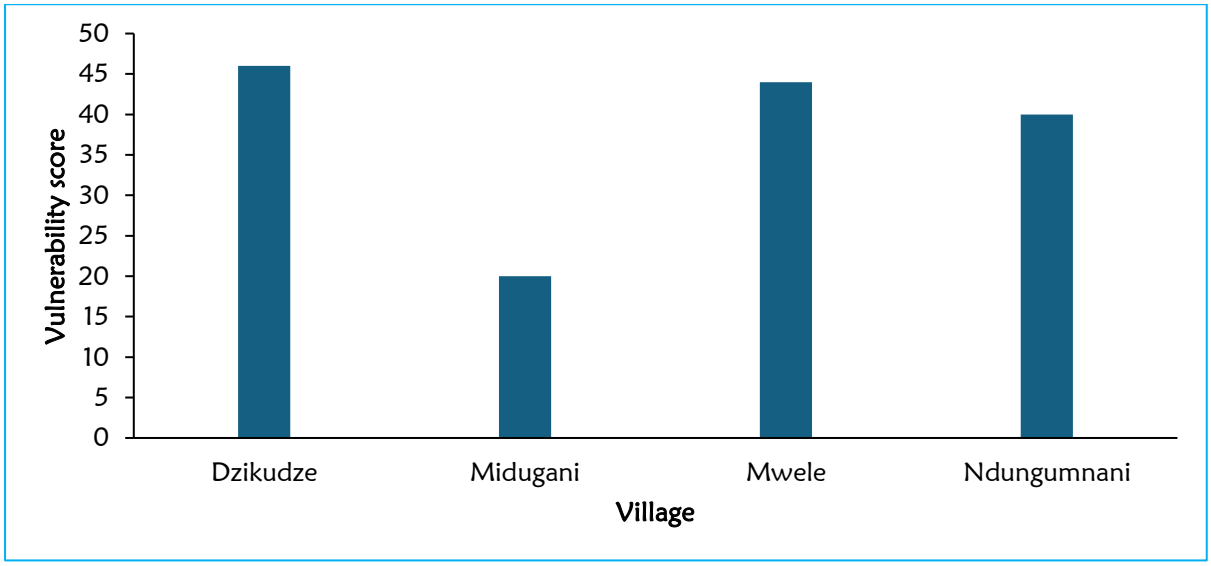

Figure 10: Drought vulnerability score in Sokoke Ward (Source: PCVA, 2025)

Natural resources serve as vital livelihood assets but face varying levels of vulnerability. Key ecosystems including Mwangea Hills, Kaya Godoma, Kaya Kadzitsoni and Arabuko Sokoke Forest, have undergone significant degradation, characterized by vegetation loss, reduced regeneration of indigenous species, and declining biodiversity. The NDVI analysis of the Mwangea Hills (Figure 9) shows distinct seasonal variations in vegetation vigor and canopy greenness influenced by changing climatic conditions. During the dry season, NDVI values range from 0.14 to 0.55, indicating moderate vegetation stress due to limited soil moisture, high temperatures, and increased evapotranspiration. In contrast, the wet season NDVI values range from 0.03 to 0.58, reflecting improved vegetation health and productivity supported by increased rainfall and improved soil water availability.Water bodies such as springs, rivers,

streams, and dams have similarly been affected, with many drying up, leading to soil infertility and a sharp decline in agricultural productivity. Consequently, communities are experiencing heightened food insecurity, livestock losses, and diminishing household incomes. These pressures have driven the adoption of unsustainable coping strategies such as sand harvesting, charcoal production, and excessive extraction of forest resources.

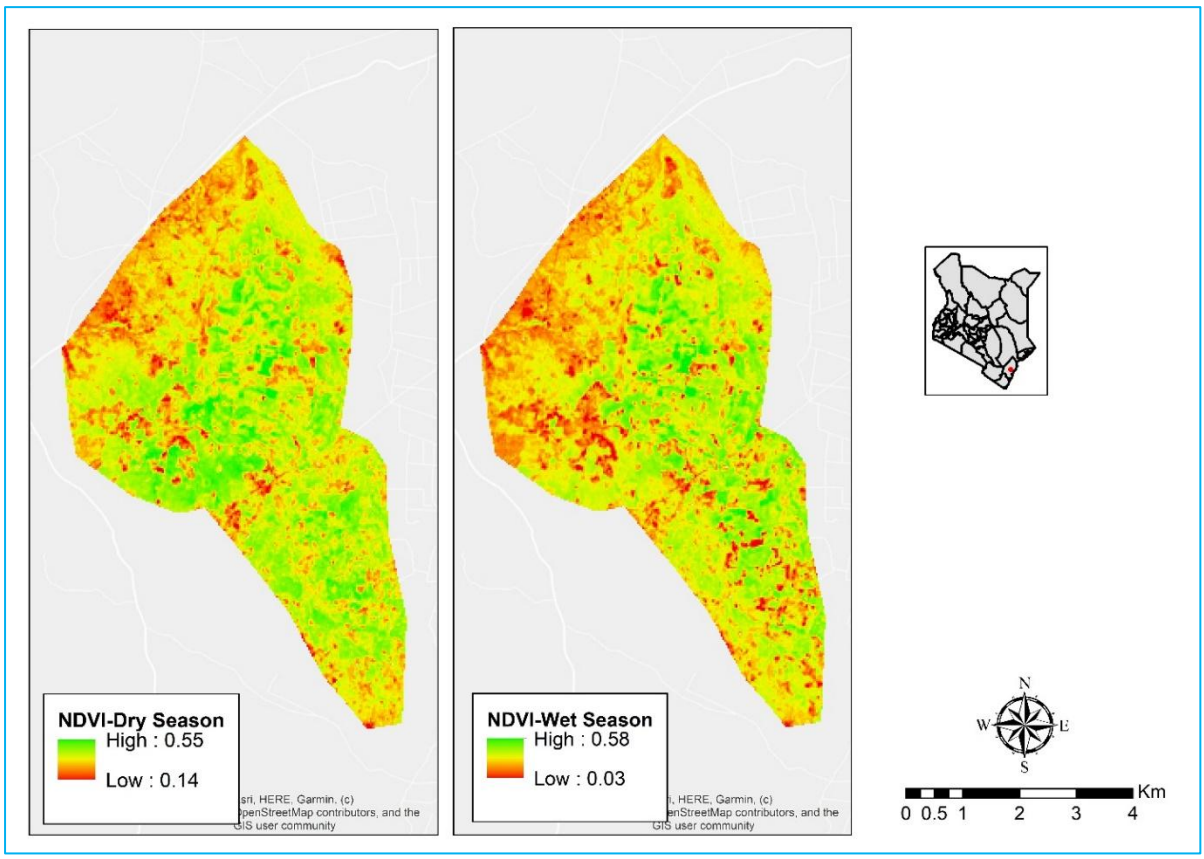

Figure 11: Mwangea Hills NDVI values across dry and wet seasons (Source: Source: Landsat 8/9 OLI and TIRS, 2024)

At the community level, drought impacts are strongly gendered and socially differentiated. Women and youth are disproportionately burdened as they bear the primary responsibility for water collection, fuelwood gathering, and household food provisioning. During prolonged drought periods, women travel long distances daily in search of potable water, reducing their time for income-generating activities and increasing exposure to heat-related health risks. Men, on the other hand, experience livelihood disruptions due to reduced agricultural productivity and loss of income from small scale farming and charcoal trade. Elderly persons and persons abled differently (PAD) face heightened vulnerability as they have limited mobility and adaptive capacity.

3.4.1.1 Exposure to drought

Exposure to drought in Sokoke Ward is extensive and cross-cutting, affecting both ecological and livelihood systems across all four villages. Key natural ecosystems, including Arabuko Sokoke Forest, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni, exhibit high exposure due to prolonged dry spells and rising temperatures. In Arabuko Sokoke Forest, persistent moisture stress has reduced canopy cover, weakened regeneration of indigenous tree species, and increased fire susceptibility threatening endemic flora and fauna that depend on the forest's microclimate. Similarly, Mwangea Hills, a vital catchment area, experiences accelerated soil erosion and drying of seasonal streams, diminishing water supply to downstream communities.

The Kaya forests (Godoma and Kadzitsoni) are also severely exposed, with declining soil moisture leading to dieback of sacred trees and loss of understorey vegetation that supports traditional biodiversity and cultural practices. As soils dry and fertility declines, farmlands surrounding these ecosystems face recurring crop failure, particularly for drought-sensitive crops such as maize. Moreover, shallow wells, boreholes and springs throughout the ward face depletion as groundwater tables recede, forcing communities to depend on unsafe or distant water sources. Livestock exposure where prolonged scarcity of pasture and water leads to increased mortality rates, reduced milk production, and recurrent livelihood losses. This wide-ranging exposure underscores the interconnectedness of drought impacts on both natural and human systems in Sokoke Ward.

3.4.1.2 Socio-Demographic Vulnerability to Drought

Vulnerability to drought in Sokoke Ward exhibits marked socio-demographic disparities shaped by gender, age, and livelihood dependence. Women, particularly those heading households, are disproportionately impacted as they shoulder the primary burden of securing water, food, and fuelwood; during prolonged dry spells, they walk longer distances to reach depleted boreholes or shallow wells or springs, heightening their physical and economic strain. Youth, especially those engaged in small scale farming, and tree nursery production experience significant livelihood disruptions as water scarcity and heat stress reduce productivity and income opportunities. Elderly persons and persons abled differently (PAD) remain highly vulnerable due to limited mobility and reliance on others for access to water and food resources. In adaptive terms, only a small fraction of households, mainly male-headed or those with diversified incomes from trade, livestock, or formal employment, are coping effectively with recurrent droughts, while the majority of smallholder farmers are declining, facing shrinking yields and deteriorating living conditions. A few households supported by ongoing interventions, such as water harvesting, demonstrate improving resilience, adopting practices that stabilize livelihoods despite the increasing drought frequency. This uneven capacity to withstand and recover from drought underscores the need for gender-responsive and socially inclusive adaptation strategies within the ward.

3.4.2 Human –wildlife conflicts

Human–wildlife conflicts (HWC) have emerged as a recurrent and significant challenge in Sokoke Ward, particularly in areas adjacent to degraded forest edges and wildlife movement corridors. The ward lies within a vital ecological linkage between Arabuko-Sokoke Forest and Tsavo East National Park, serving as a key dispersal and migration zone for wildlife species. Consequently analysis of data from 2017 to 2025 shows that interactions between people and wildlife have become more frequent, occurring roughly once every year (Figure 10). Spatial analysis indicates that Dkunze records the highest vulnerability (score 45), followed by Midugani (13) and Ndungumnani (11) (Figure 11). This spatial variation is influenced by proximity to forest boundaries, land-use intensity, and the distribution of water and forage resources.

The degradation of Mwangea Hills, Arabuko Sokoke Forest, Kaya Godoma, and Kaya Kadzitsoni has significantly heightened exposure to human–wildlife conflicts in Sokoke Ward. The loss of forest cover, depletion of water sources, and reduction of natural forage have disrupted traditional wildlife habitats and migratory patterns, forcing animals to move into nearby farmlands and settlements in search of food and water. In Dzikudze Village, vulnerability analysis indicates the highest overall score (45), with crop farming (score 5) and livestock keeping (score 5) identified as the most impacted livelihood sectors. Elephants are known to trample and destroy standing crops such as maize and cassava; while baboons and monkeys raid farms, leading to frequent yield losses and heightened food insecurity. Additionally,

livestock particularly goats and poultry occasionally fall prey to carnivores, compounding economic losses for smallholder farmers.

From a socio-demographic perspective, farmers living along the Forest edges and pastoralists in open grazing areas are the most exposed groups. Women and children, who spend long hours on farms and at water collection points, face heightened physical danger from wildlife encounters, while widows, elderly persons, and persons abled differently (PAD) are particularly vulnerable due to limited mobility and inability to construct deterrent fences or invest in alternative livelihoods. The destruction of crops and livestock not only threatens household food security but also forces families to depend on purchased food or relief support, thereby deepening poverty levels. Over time, HWC incidents have expanded in both frequency and severity, reflecting a broader ecological imbalance as wildlife habitats shrink and human settlements expand.

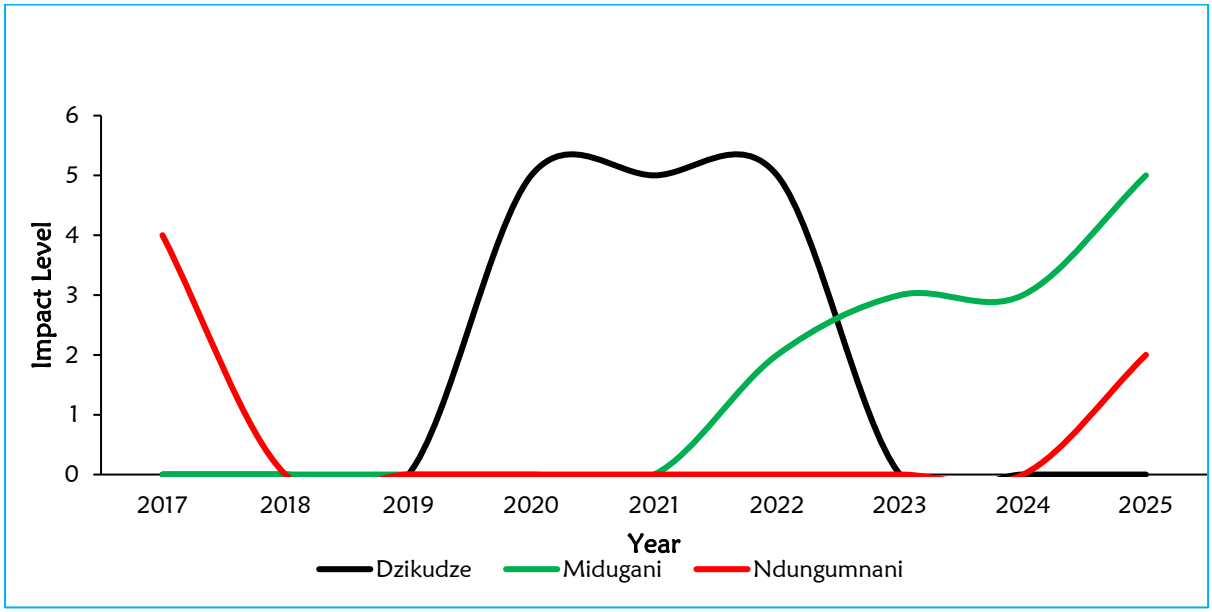

Figure 12: Human-wildlife prevalence historical trend (Source: PCVA, 2025)

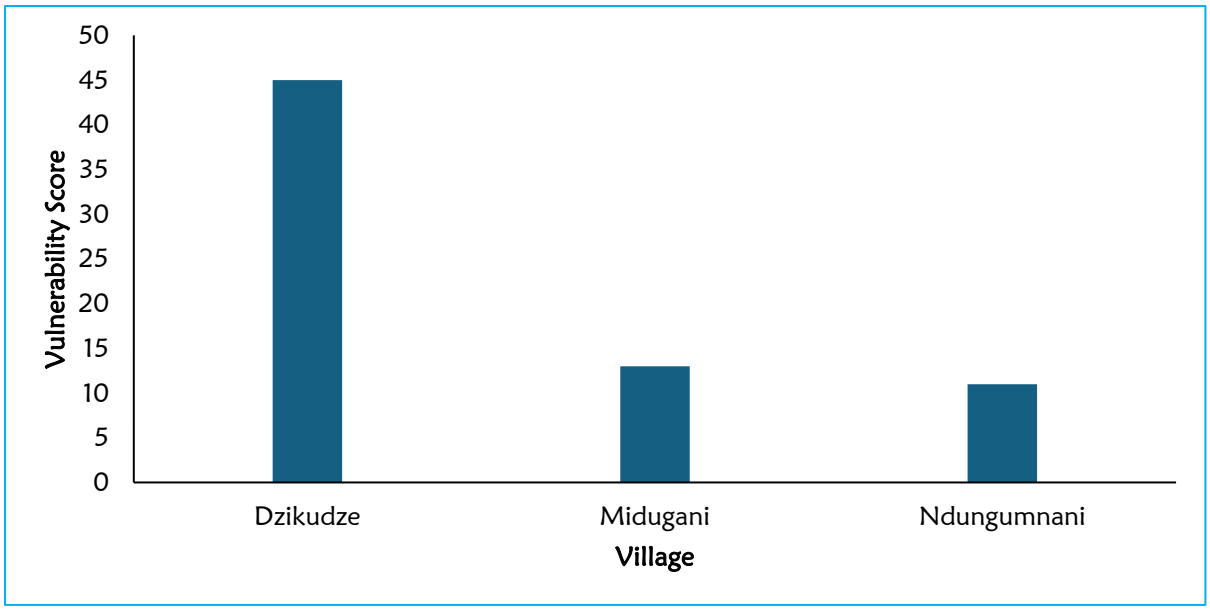

Figure 13: Human-wildlife vulnerability score in Sokoke Ward (Source: PCVA, 2025)

3.4.2.1 Exposure to Human-Wildlife Conflicts

Exposure to human–wildlife conflicts in Sokoke Ward is most pronounced in communities situated near forested and degraded habitats, particularly around Dkunze Village, which borders the Arabuko Sokoke Forest. These areas experience frequent interactions between humans and wildlife due to declining vegetation cover, drying water sources, and the continued expansion of farmlands into former forest buffer zones. The degradation of Arabuko Sokoke Forest, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni has significantly weakened natural barriers and reduced food availability for wildlife, forcing species such as elephants, baboons, and monkeys to move into cultivated areas in search of forage. This has led to widespread destruction of crops, fruit trees, and young seedlings particularly during dry seasons when natural food sources are limited. Livestock keepers grazing near forest fringes are also exposed to predation and resource competition. Women and children, who often spend long hours in farms, near forest edges, or collecting water and firewood, face the highest physical risk from wildlife encounters. Meanwhile, widows, the elderly, and persons abled differently (PAD) are disproportionately vulnerable due to limited mobility, restricted access to protective measures, and reduced capacity to respond quickly to wildlife threats.

3.4.2.2 Socio-Demographic Vulnerability to Human -Wildlife Conflicts

Socio-demographic vulnerability to human–wildlife conflicts in Sokoke Ward is shaped by livelihood type, gender, age, and proximity to wildlife habitats, with the highest vulnerability observed among resource-dependent and low-income households. Smallholder farmers and pastoralists residing near Arabuko Sokoke Forest are the most affected, as their livelihoods rely heavily on land and water resources that overlap with wildlife ranges. Frequent crop raids by elephants, monkeys, and baboons, and predation of goats and poultry, undermine household food security and income stability. These losses are especially devastating for poor households that lack access to compensation mechanisms or alternative livelihood options.

Women are disproportionately vulnerable because they perform daily activities that heighten their exposure, including farming, firewood collection, and water fetching near forested areas. The constant fear of wildlife encounters limits their mobility and reduces productivity in agriculture and trade. Children also face high vulnerability as they often accompany parents to farms or water sources, exposing them to physical danger. Widows, elderly persons, and persons abled differently (PAD) experience compounded vulnerability due to restricted mobility, limited ability to construct deterrent barriers, and dependence on external assistance for protection and recovery after incidents. The economic ripple effects of HWCs also intensify vulnerability across the broader community. Destruction of crops and livestock forces households to purchase food, depend on relief aid, or migrate temporarily, disrupting education, social cohesion, and local markets. Over time, the frequency and severity of HWCs have deepened social inequalities, as wealthier households with stronger structures, fences, or alternative income sources recover faster, while poorer families remain trapped in cycles of loss and indebtedness. Overall, socio-demographic vulnerability to HWCs in Sokoke Ward highlights the intersection of environmental degradation, poverty, and gender inequality, emphasizing the need for inclusive wildlife management strategies, gender-responsive compensation schemes, and community-led mitigation measures that protect the most at-risk groups while promoting coexistence between people and wildlife.

3.4.3 Environmental Degradation

Environmental degradation in Sokoke Ward is a growing hazard driven largely by unsustainable human activities such as tree felling for charcoal production, timber, and fuelwood. Analysis of historical trends between 1987 and 2025 reveals that major degradation episodes have occurred approximately once every five (5) years, with notable peaks in 1997 and 2024 corresponding to periods of rapid population growth, expansion of settlements, and intensified

agricultural encroachment, particularly around Dzikudze Village (Figure 12). Over time, the frequency and severity of degradation have increased, transforming once localized disturbances into extensive ecosystem decline affecting forests, riverbanks, wetlands, and coastal zones. Vulnerability assessments show that Dzikudze exhibits the highest vulnerability score (41), followed by Mwele (35), reflecting the extent of ecological stress in these areas (Figure 13).

The Arabuko Sokoke Forest, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni are among the most severely affected terrestrial ecosystems, having suffered extensive tree felling for fuelwood, charcoal, and construction materials. The resulting loss of forest cover has disrupted local hydrological processes, reduced groundwater recharge, and increased surface runoff intensifying both drought and flood risks. This degradation has also led to a steady decline in biodiversity and weakened critical ecosystem services such as water regulation, carbon sequestration, and soil stabilization. Consequently, the deterioration of these ecosystems not only amplifies community vulnerability to climate-related hazards but also aggravates human–wildlife conflicts, as animals are forced to move into nearby settlements and farmlands in search of food, water, and shelter.

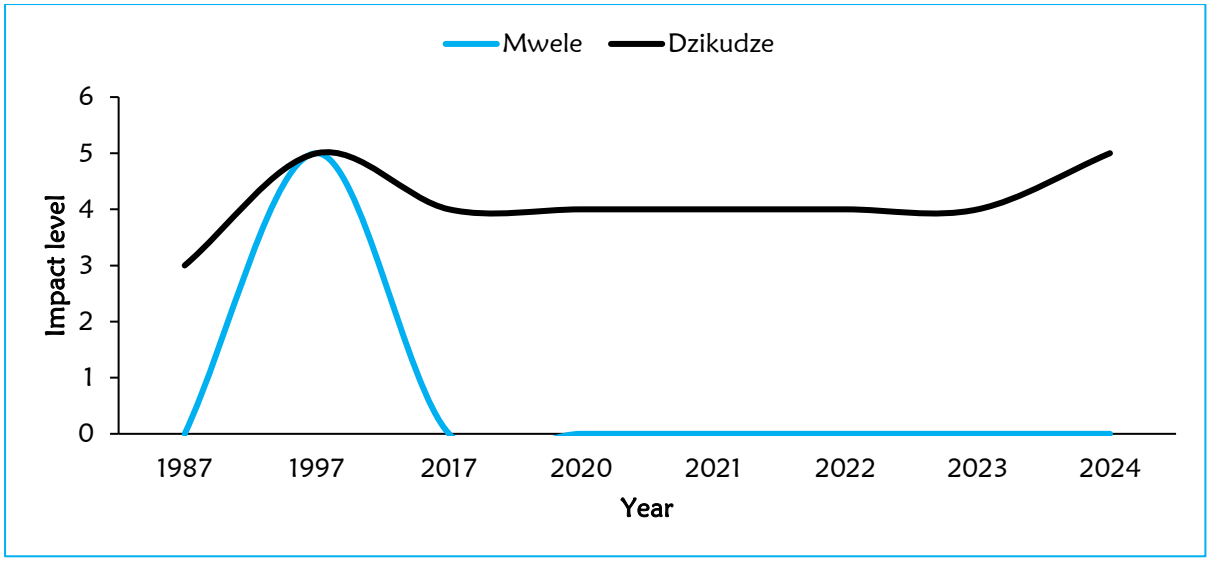

Figure 14: Environmental Degradation prevalence historical trend (Source: PCVA, 2025)

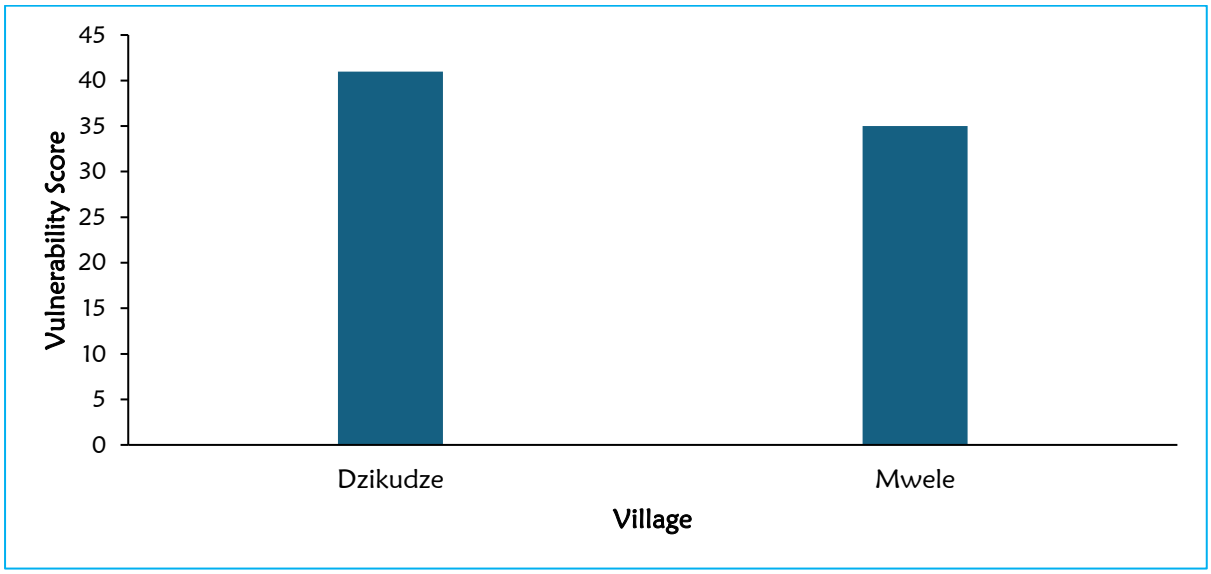

Figure 15: Environmental Degradation vulnerability score in Sokoke Ward (Source: PCVA, 2025)

3.4.3.1 Exposure to Environmental Degradation

Exposure to environmental degradation in Sokoke Ward is extensive, affecting both terrestrial ecosystems and the communities whose livelihoods depend on them. The most impacted areas include Arabuko Sokoke Forest, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni, where unregulated human activities have disrupted ecological balance and reduced ecosystem productivity. Widespread clearing of vegetation for cultivation, fuelwood extraction, and settlement expansion has left large tracts of land exposed to erosion, diminished soil fertility, and altered local hydrological cycles. The degradation of the Arabuko Sokoke Forest, in particular, has not only accelerated biodiversity loss but also heightened community vulnerability to floods and droughts due to reduced groundwater recharge and infiltration capacity.

Communities in Dzikudze and Mwele villages face the greatest exposure, given their direct dependence on these ecosystems for food, water, fuel, and income. Women and youth, who rely heavily on forest resources and rain-fed farming, are disproportionately affected as they must travel longer distances to collect water, firewood, and other essentials intensifying both physical strain and economic vulnerability. Conversely, households with more diversified income sources exhibit moderate exposure, as their dependence on natural ecosystems for daily sustenance is comparatively lower.

3.4.3.2 Socio-Demographic Vulnerability to Environmental Degradation

Socio-demographic vulnerability to environmental degradation in Sokoke Ward is shaped by gender roles, livelihood dependence, income disparities, and unequal access to natural resources. The most affected groups include women, smallholder farmers, and youth, whose livelihoods are directly tied to forest ecosystems for food, fuel, and income. Women are particularly burdened, as they rely heavily on forest resources such as fuelwood, wild fruits, and medicinal herbs. With escalating deforestation in the Arabuko Sokoke Forest, many women now walk over five kilometres daily in search of firewood and other essentials. This prolonged effort not only exposes them to physical exhaustion and safety risks but also limits time for childcare, education, and income-generating activities thereby deepening existing gender inequalities.

Smallholder farmers are equally vulnerable due to declining soil fertility, loss of vegetation cover, and intensified erosion in upland zones. The degradation of surrounding forest ecosystems has disrupted natural water cycles, reducing crop productivity and groundwater recharge, which undermines both food security and household resilience. Youth, on the other hand, increasingly rely on unsustainable activities such as charcoal production and sand harvesting for income livelihoods that accelerate ecosystem degradation and perpetuate economic instability. Meanwhile, elderly persons and persons abled differently (PAD) face further exclusion, as the depletion of nearby natural resources constrains their access to essential goods like firewood, clean water, and traditional medicinal plants.

Spatially, the highest socio-demographic vulnerability is concentrated in Dzikudze and Mwele villages, where forest loss is most severe and livelihood alternatives are scarce. Communities here are trapped in a poverty degradation cycle, where resource depletion reduces productivity and poverty, in turn, drives unsustainable extraction. Only a few moderately resilient households typically those with diversified incomes from trade, formal employment, or remittances exhibit improved adaptive capacity through reduced dependence on natural ecosystems. Overall, environmental degradation in Sokoke Ward has deepened existing social inequities, with women, the poor, and resource-dependent groups bearing the greatest vulnerability. This underscores the urgent need for gender-responsive restoration programs,

sustainable livelihood diversification, and community-led ecosystem management to rebuild both ecological and social resilience.

3.4.1 Pests and Diseases

Incidences of pests and diseases have persistently affected Sokoke Ward, posing significant threats to livelihoods and food security. Historical data from 2011 to 2025 indicate that outbreaks occur approximately every two years, with notable peaks in 2011, 2019, and 2025 (Figure 14). Over time, both the frequency and spatial extent of these outbreaks have intensified, largely driven by erratic rainfall patterns, rising temperatures, and prolonged dry spells that weaken crop and livestock resistance. Findings from the Participatory Climate Vulnerability Assessment (PCVA) reveal that Mwele is the most affected area (vulnerability score 37), followed by Midugani (score 16) (Figure 15).

Natural ecosystems and critical livelihood resources have been equally compromised. The Arabuko Sokoke Forest, Kaya Godoma, and Kaya Kadzitsoni key ecological assets that provide vital ecosystem services such as soil stabilization, water regulation, and microclimate moderation have increasingly suffered from pest-related leaf defoliation and fungal infections. These disturbances have impaired forest regeneration, reduced biodiversity, and diminished the ecological resilience of the ward. Beyond forest degradation, several other natural and livelihood systems have also been adversely affected. Agricultural systems, particularly maize, cassava, and cowpeas, have experienced recurrent pest infestations such as armyworms, aphids, and whiteflies, leading to significant yield declines.

At the community level, exposure to pest and disease risks is shaped by socio-economic and gender dynamics. Smallholder farmers, whose livelihoods depend heavily on climate-sensitive resources, remain the most exposed group, as pest infestations and livestock diseases directly erode their productivity and income. Women, who constitute the majority of subsistence farmers, experience disproportionate impacts through reduced harvests, increased replanting costs, and heavier labour burdens associated with pest management. Pastoral households are similarly affected, with recurrent livestock diseases such as foot-and-mouth and tick-borne infections intensifying during wetter periods that often follow extended droughts.

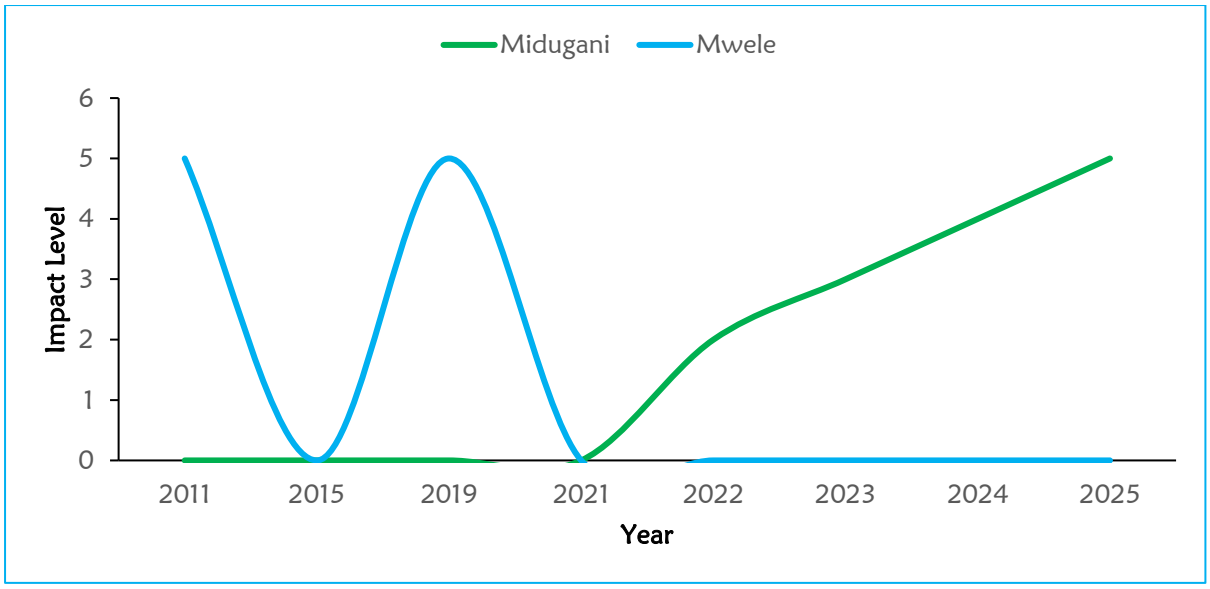

Figure 16: Pests and diseases prevalence historical trend (Source: PCVA, *2025*)

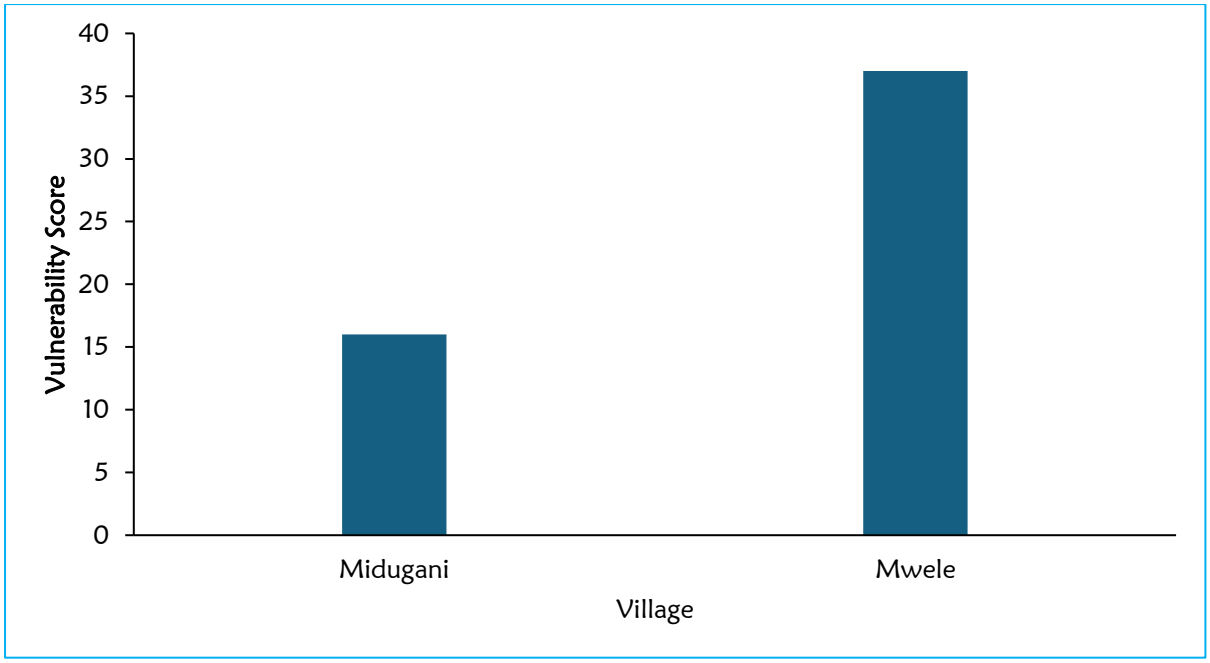

Figure 17: Pests and diseases vulnerability score in Sokoke Ward (Source: PCVA, 2025)

3.4.1.1 Exposure to Pests and Diseases

Exposure to pests and diseases in Sokoke Ward is widespread, affecting all villages and livelihood systems across the terrestrial environments. The entire ward faces recurrent infestations and outbreaks that impact crops, livestock, and human health due to changing climatic conditions, particularly fluctuations in temperature, rainfall, and humidity. Periodic pest invasions such as locusts, armyworms, have devastated farmlands in Midugani and Mwele, while livestock diseases such as foot-and-mouth and tick-borne infections have increased in frequency following alternating drought and wet seasons. This exposure is heightened by the ward's dependence on rain-fed agriculture, open grazing systems, and the proximity of settlements to degraded forest and wetland habitats that harbour pest breeding sites.

3.4.1.2 Socio-Demographic Vulnerability to Pests and Diseases

Socio-demographic vulnerability to pests and diseases in Sokoke Ward is deeply influenced by livelihood dependence, gender roles, age, and access to adaptive resources. The most affected groups are smallholder farmers and pastoralists, whose livelihoods are directly dependent on climate-sensitive natural resources. Women, who form the majority of these groups, are particularly vulnerable as they manage subsistence farms and tend small livestock that are highly susceptible to pest and disease outbreaks. When infestations such as locusts and armyworms, destroy crops, women's income and household food security are severely compromised. Their adaptive capacity is further constrained by limited access to pest control technologies, extension services, and credit facilities, forcing them to rely on traditional coping mechanisms that are often ineffective under escalating climate pressures.

Youth face heightened vulnerability due to their reliance on casual labour and low-paying agricultural livelihoods that are disrupted during pest and disease outbreaks. The loss of these income sources leads to increased unemployment and migration pressures. Elderly persons and persons abled differently (PAD) are also at high risk, as physical limitations restrict their ability to engage in pest management activities or relocate when infestations and disease outbreaks intensify. In Midugani and Mwele, where exposure levels are highest, livestock diseases such as foot-and-mouth and tick-borne infections repeatedly erode household assets, disproportionately affecting poorer families who lack veterinary support or the capacity to restock.

From a health perspective, children and elderly individuals are more susceptible to secondary human disease outbreaks, such as malaria, dengue fever, and chikungunya illnesses, that often accompany climatic shifts and pest proliferation. In contrast, a minority of better-off households, particularly those with diversified income sources or access to agricultural extension services, exhibit moderate coping capacity through the use of improved seeds and pesticides. However, the majority of vulnerable groups remain declining, with repeated losses eroding resilience over time. Overall, socio-demographic vulnerability to pests and diseases in Sokoke Ward underscores entrenched gender and economic inequalities, emphasizing the need for inclusive adaptation measures that enhance women's access to information, promote climate-smart livelihoods, and strengthen community-based pest and disease surveillance systems.

3.4.2 Floods

The ward rarely experiences flooding due to its sandy soils, which allow rapid water infiltration. Historical data from 1961 to 2025 indicate that outbreaks occur approximately every 13 years, with notable peaks in 1961, 1997, and 1998 (Figure 16). Vulnerability analysis shows that flooding is mostly experienced in Mwele village (Figure 17). The effects of floods on natural and livelihood resources are widespread and multifaceted. Arabuko Sokoke Forest, Kaya Godoma, and Kaya Kadzitsoni experience soil erosion, excessive surface runoff, and loss of ground cover, which undermines their ecological functions in soil stabilization, groundwater recharge, and hydrological regulation.

Flooding severely disrupts community life and disproportionately affects vulnerable groups. Women face the greatest burden as floods destroy water sources, roads, and sanitation facilities, forcing them to spend more time fetching water, caring for children and the sick, and managing damaged homes greatly increasing unpaid care work and reducing time for income-generating activities. Youth and men are often involved in immediate response activities including assisting households to relocate, or salvaging belongings. These efforts, though vital, divert time from daily wage labour, reducing household income and increasing economic vulnerability. Elderly persons and people with disabilities often struggle to access relief and evacuation centres due to mobility challenges, leaving them highly dependent on others for support. Prolonged flooding also cuts off markets, schools, and health facilities, deepening food insecurity and straining household resilience.

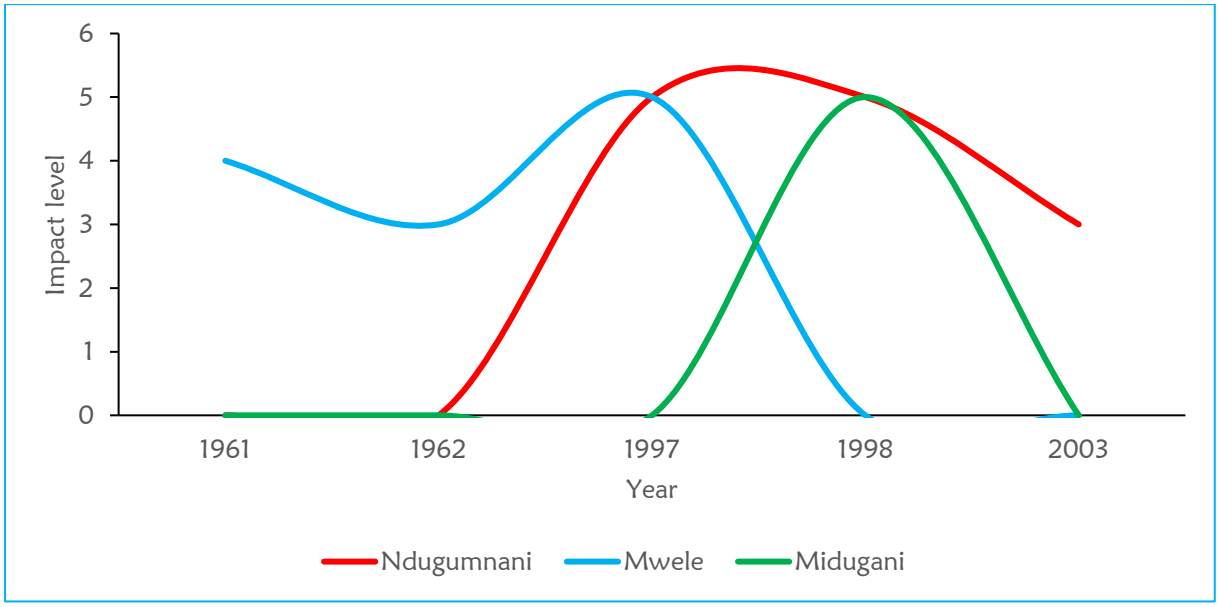

Figure 18: Floods prevalence historical trend (Source: PCVA, 2025)

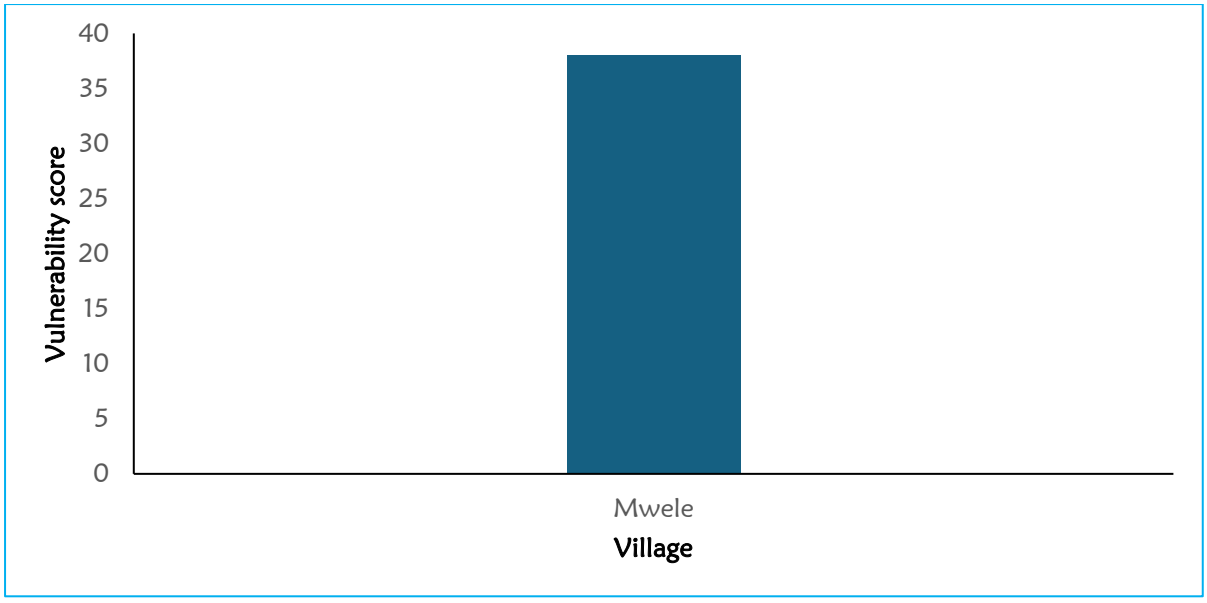

Figure 19: Floods vulnerability score in Sokoke Ward (Source: PCVA, 2025)

3.4.2.1 Exposure to Floods

Flood exposure in Sokoke Ward is widespread and unevenly distributed, affecting both terrestrial ecosystems and human settlements. The most flood-prone areas occur in low-lying and riparian zones along rivers and streams such as Milore, Rare, Muriachakwe, Katangani, and Nyare. Recurrent flooding in these areas causes destruction of croplands, loss of fertile topsoil, and erosion of access roads, disrupting mobility and isolating communities during peak rainfall periods. In low-lying settlements located near drainage channels, inadequate stormwater infrastructure worsens surface flooding, leading to repeated property damage and temporary displacement of residents. The convergence of these exposure hotspots reflects a complex interplay between altered hydrological patterns, land-use degradation, and unregulated settlement expansion. Collectively, these factors heighten the vulnerability of socio-economically marginalized and resource-dependent households across Sokoke Ward.

3.4.2.2 Socio-Demographic Vulnerability to Floods

Socio-demographic vulnerability to floods in Sokoke Ward is shaped by gender, age, physical ability, livelihood type, and socio-economic status, all of which influence how different groups experience, respond to, and recover from flood events. Women, particularly those heading households or engaged in informal trade, are disproportionately affected due to limited access to productive assets and their dual domestic and economic responsibilities. During flood periods, they shoulder additional burdens such as caring for sick family members, fetching safe water from distant sources after wells become contaminated, and managing food shortages when access to markets and farmlands is disrupted. Children are similarly at high risk owing to exposure to waterborne diseases such as cholera and diarrhoea, unsafe play in flooded environments, and interruption of schooling when learning institutions are submerged or used as temporary shelters. Elderly persons and persons abled differently (PAD) face unique challenges related to mobility, evacuation, and access to medical services, making them more dependent on others for survival during emergencies.

Economically, low-income households, especially those residing near flood-prone zones are least able to recover from flood-induced losses due to limited savings and inadequate access to insurance or credit facilities. Their housing structures, often made of mud walls and thatched roofs, are easily destroyed, forcing repeated rebuilding that deepens poverty cycles. Youth engaged in casual labour, or small-scale farming also experience recurring income shocks as floods disrupt agricultural production, further undermining household resilience. Conversely, a

small segment of households with diversified income sources, such as salaried employment or established trade, demonstrate moderate coping ability and quicker recovery after flood events.

3.4.3 Whirl Wind

Whirlwinds have emerged as a recurrent climate hazard in Sokoke Ward, causing significant damage to property and vegetation by blowing off house roofs, uprooting trees, and destroying farmlands. Historical records between 2023 and 2025 indicate that these events occur almost annually, with a major outbreak recorded in 2024 (Figure 18). Vulnerability analysis identifies Ndugumnani Village as the most affected area (Figure 19).While whirlwinds are reported across the entire ward, their impacts are most pronounced in low-lying and open landscapes such as Mwele, Dzikudze, Ndigiria, Mapotea, Mitedzini, Mrima wa Ndege, and surrounding villages. The sparse vegetation cover and exposure to strong wind corridors in these areas increase the likelihood of structural damage and soil erosion. Poorly constructed houses with lightweight roofing materials are particularly vulnerable, compounding the risk for low-income households.

Whirlwinds have significantly affected natural ecosystems and critical livelihood resources in Sokoke Ward. The strong gusts uproot trees, strip leaves, and break branches in forested areas such as the Arabuko Sokoke Forest, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni, thereby disrupting canopy structure and reducing habitat integrity for birds and other wildlife. The destruction of tree cover accelerates soil erosion, decreases organic matter accumulation, and undermines natural regeneration processes. In farmlands, whirlwinds flatten crops such as maize, cassava, and coconut, leading to direct yield losses and reduced food security. Livelihood assets, including granaries, livestock shelters, and beehives, are often destroyed, resulting in economic losses for households that depend on small-scale farming and apiculture. Over time, the repeated disturbance of vegetation and soil surfaces also reduces the ecological resilience of the ward, diminishing its capacity to recover from subsequent climatic shocks such as droughts and floods.

From a community perspective are increasingly destructive hazard that causes sudden and unpredictable damage to homes, farmlands, and public infrastructure. Women, who are primarily responsible for household management and small-scale farming, bear the greatest burden. They face direct losses when food stores, kitchens, or drying structures are destroyed, and must also invest additional labour in replanting and rebuilding. The loss of roofing materials and household items places financial strain on already limited incomes, deepening their economic vulnerability. Youth and men are often at the frontline of recovery, tasked with clearing fallen trees, repairing damaged houses, and assisting vulnerable community members. However, limited access to financial resources and employment opportunities constrains their ability to rebuild resilient structures, leaving them trapped in a cycle of reconstruction and loss. Elderly persons and PADs face heightened physical and mobility challenges during and after whirlwind events. They are often unable to seek shelter quickly or participate effectively in rebuilding efforts, making them dependent on family and community support.

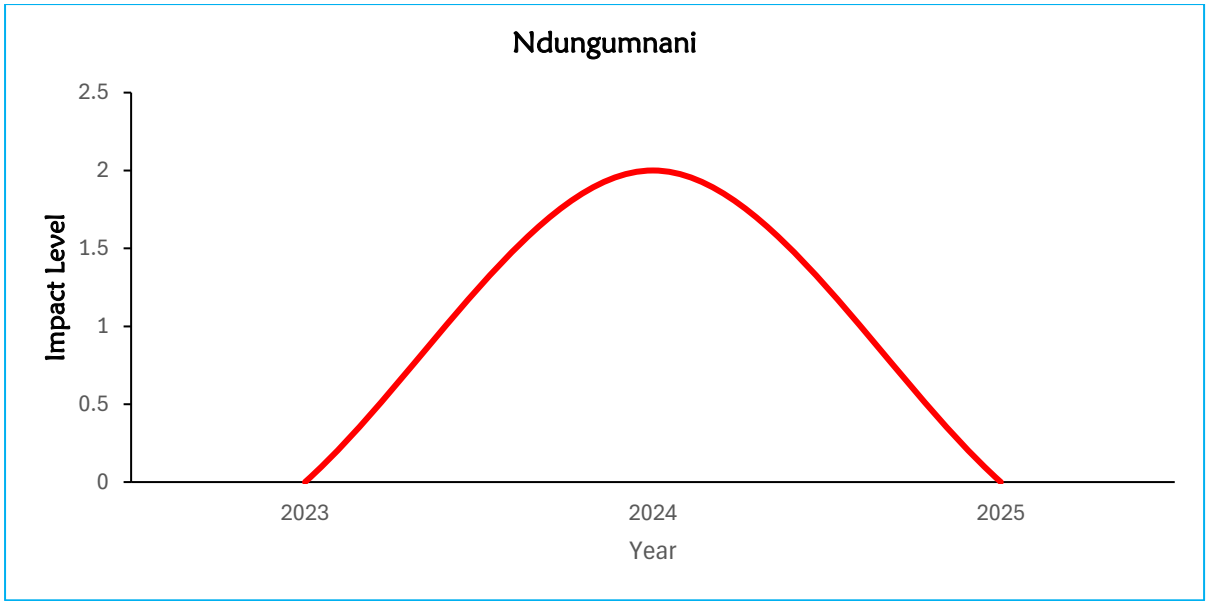

Figure 20: Whirlwinds prevalence historical trend (Source: PCVA, 2025)

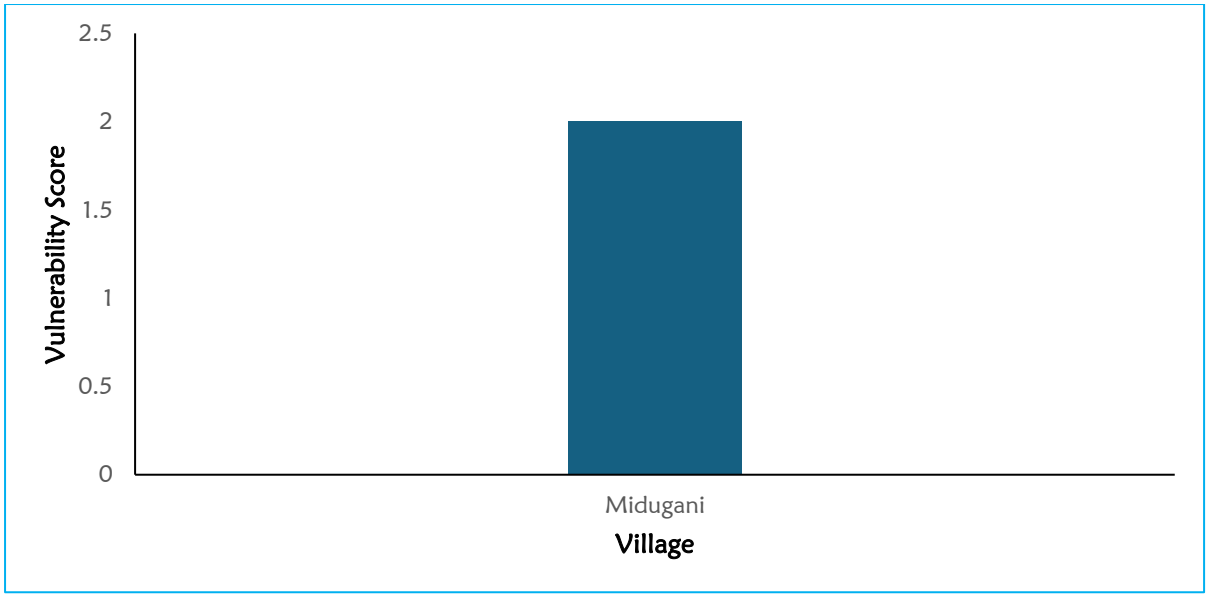

Figure 21: Whirl Wind vulnerability score in Sokoke Ward (Source: PCVA, 2025)

3.4.3.1 Exposure to Whirl wind

Exposure to whirlwinds in Sokoke Ward is widespread, with impacts felt across all villages, particularly in open and low-lying areas where vegetation cover is sparse and housing structures are less resilient. The most affected settlements include Mwele, Ndugumnani, Dzikunze, Midzedzini, and Mrima wa Ndege, where residents reported frequent occurrences of strong winds capable of blowing off roofs, uprooting trees, and damaging household structures. Farmlands located on exposed plains are especially vulnerable, as whirlwinds destroy crops, young tree seedlings, and fruit trees, leading to loss of both food and income. The situation is aggravated by widespread deforestation and land clearing for agriculture, which have reduced natural windbreaks such as trees and shrubs. Schools, markets, and community buildings with poorly maintained roofing also face recurrent structural damage, disrupting learning and community activities. The exposure of households is closely linked to poverty levels and housing conditions with most homes built using lightweight materials such as makuti, iron sheets, and

mud walls that are easily damaged by strong winds. In most villages, residents lack access to affordable building materials and technical support for constructing wind-resistant structures.

3.4.3.2 Socio-Demographic Vulnerability to Whirl Winds

Socio-demographic vulnerability to whirlwinds in Sokoke Ward varies across different social groups and is shaped by gender roles, age, livelihood dependence, and access to resources. Women, youth, the elderly, and persons abled differently (PAD) are among the most affected due to limited mobility, lower adaptive capacity, and exposure linked to daily livelihood activities. Women are particularly vulnerable as they spend more time around homesteads, markets, and farmlands where housing and storage structures are often weak. During whirlwind events, they face heightened risks of injury and property loss, especially as they care for children and elderly family members. The destruction of food stores and household assets further compounds their economic vulnerability, increasing dependence on borrowed resources or humanitarian support.

Youth, who often engage in casual work such as construction or vending in open-air markets, face exposure through occupational risks. Damage to markets, roads, and small enterprises reduces their income and employment opportunities, pushing many to migrate temporarily in search of alternative livelihoods. Elderly persons and PAD are highly vulnerable due to limited mobility and difficulty in evacuating or seeking shelter during strong wind events. Many reside in older or poorly maintained houses that are more susceptible to roof damage and collapse. Following such events, they often lack the physical and financial means to repair their homes or recover lost belongings, deepening their dependence on family or community assistance.

3.5 Classification of Exposure and Resilience through Livelihoods

The ward is characterised by livelihoods that are highly exposed to environmental, economic and institutional shocks, while its capacity for resilience remains constrained by limited income diversity, low margins and persistent poverty. The dominant cash-income sources are informal and closely tied to transport services with modest supplementary incomes from micro-enterprises, small-scale agriculture and livestock or poultry. At the same time, county data indicate that Kilifi's poverty headcount stands at approximately 46.1 percent, which is notably higher than the national average, reflecting widespread income deprivation and limited livelihood diversification across rural areas. These factors combine to yield a system with narrow buffers and high vulnerability, thereby demanding careful classification of exposure and resilience through the livelihood lens.

3.5.1 Livelihood Structure and Monthly Incomes

Table 2 below presents differentiated typical net monthly earnings (KES) per actively engaged adult in each of the four villages (Dzikudza, Ndugumnani, Mwele and Midugani).The findings indicate variations in household income across the four villages, reflecting differences in resource access, proximity to ecosystems, and exposure to climate and environmental risks. Livelihoods in the area are intricately linked to natural ecosystems such as Arabuko-Sokoke Forest, Mwangea Hill, Kaya Godoma, and Kaya Kadzitsoni, which provide essential resources, regulate local microclimates, and sustain ecosystem services that underpin local economies. The dominant livelihood sources include sand harvesting, charcoal burning, and motorbike transport, which form the upper-income tier, with monthly returns ranging from KES 48,000, 30,000, and 24,000 respectively. In contrast agriculture, casual labour, and firewood sales represent low-income and more vulnerable livelihood options.

Sand harvesting stands out as one of the most lucrative livelihood activities within the ward, particularly for men and youth who supply the growing construction market in nearby towns. The activity provides immediate cash income and supports a network of transporters, loaders,

and traders, making it a central pillar in the local informal economy. Its profitability is driven by strong and steady demand for building materials, often yielding daily earnings that far exceed those from small-scale agriculture or trade. This has positioned sand harvesters as key economic contributors within their communities, sustaining household needs such as food, school fees, and healthcare.

Charcoal burning remains a significant livelihood activity for many low-income households, particularly those residing near forested landscapes such as Arabuko-Sokoke, Mwangea Hill, and the Kaya forests. It offers an accessible source of income with minimal start-up costs and provides a crucial financial buffer during agricultural downturns. The trade supports a small but active value chain involving producers, transporters, and vendors supplying both local and urban energy markets. However, the practice is inherently unsustainable when unregulated, as it relies heavily on indigenous tree species, leading to deforestation, habitat loss, and soil degradation. The resulting decline in forest cover disrupts watershed functions and microclimatic balance, undermining the ecological integrity of critical ecosystems and the long-term viability of other nature-dependent livelihoods such as beekeeping, herbal collection, and farming

The high returns from motorbike operations are largely attributed to their central role in daily household logistics and community mobility. Most residents rely on motorbikes to transport children to and from school and to access markets, health facilities, and social amenities. On average, households spend about KES 200 per day on school transport alone, creating a steady and reliable demand for motorbike services. This dependence has positioned motorbike operators as critical economic actors within the ward, contributing to consistent income flows despite periodic shocks such as fluctuating fuel prices, poor road infrastructure, and mechanical breakdowns.

In contrast, low-income livelihoods such as subsistence agriculture, firewood sale, casual labour, water vending, small-scale trade, and the sale of herbs generate modest and often irregular monthly earnings, typically ranging between KES 2,000 and KES 10,000. These activities are primarily undertaken by women, youth, and vulnerable households with limited access to land, capital, or markets. Subsistence agriculture ensures household food security but contributes little cash income, while firewood collection, water vending, and herb sales depend on local forests such as Arabuko-Sokoke and Mwangea Hill. Casual labour and small-scale trade provide short-term income but are highly sensitive to seasonal demand. Overall, these livelihoods form the economic base for many households but remain precarious, offering limited savings potential or resilience to shocks.

Table 2: Monthly earnings by livelihood and village (Source: PCVA, 2025)

Livelihood	Income (KES)			
	Ndugumnani	Mwele	Dzikudze	Midugani
Agriculture	3,000	10,000	2,000	2,500
Charcoal burning	30,000	-	4,500	30,000
Firewood sale	-	-	3,000	16,000
Livestock production	8,300	2,500	-	2,500
Motorbike transport	-	-	10,000	24,000
Small scale business	-	-	8,000	10,000
Sand harvesting	-	48,000	-	-
Sale of herbs	1,600	-	-	-
Casual labour	-	-	3,500	-
Water vending	-	-	3,000	-

The poverty profile of the four villages reflects the fragility of livelihoods, many of which rely on natural resources that are increasingly depleted. Field data show that a significant number of households in Ndugumnani, Mwele, Dzikudze, and Midugani went without sufficient income over the past year to meet basic needs such as food, housing, clothing, transport, and medical care (Figure 20).

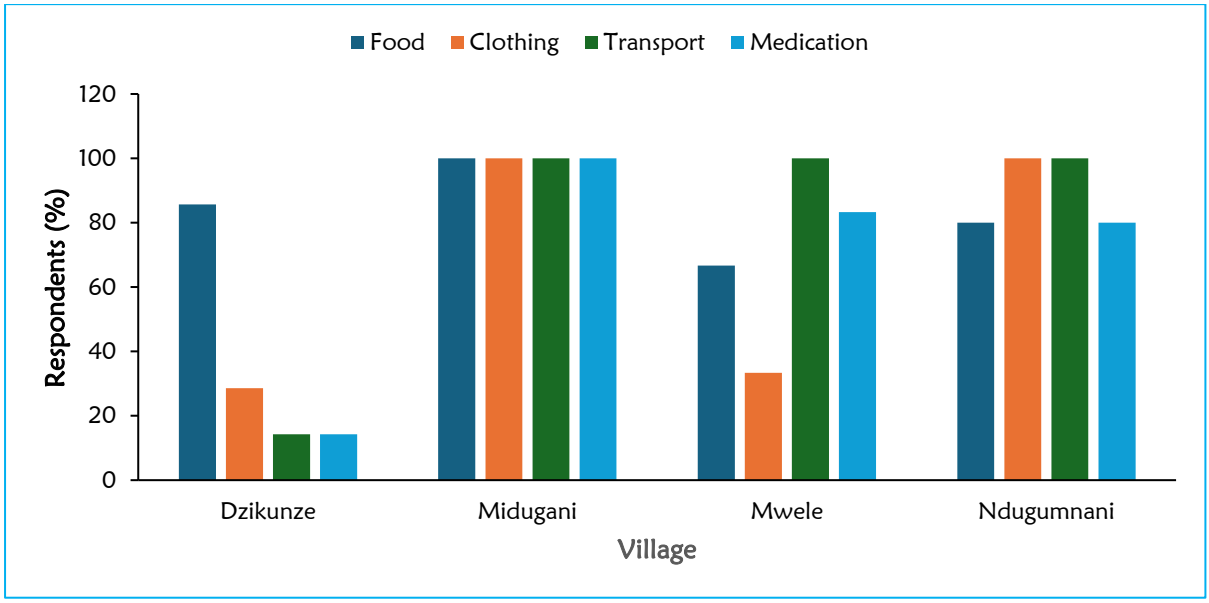

Figure 22: Respondents who lacked money to Meet Basic Needs in the Last 12 Months (Source: PCVA 2025)

The data indicate widespread deprivation across the ward, though the severity varies by village and need. Food deprivation ranges from approximately 67% in Mwele to 100% in Midugani, while clothing shortages affect 30% of households in Dzikunze up to 100% in Midugani and Ndugumnani. Transport deprivation is relatively low in Dzikunze (15%) but universal in Midugani, Mwele, and Ndugumnani. Access to medication is similarly constrained, ranging from 15% of households in Dzikunze to full deprivation in Midugani. These variations highlight that while deprivation is prevalent, its intensity differs across villages, reflecting localized vulnerabilities and resource gaps.

Poverty in Sokoke Ward thus manifests as a chronic, multidimensional phenomenon rather than a temporary financial short-fall. According to the 2023–2027 Kilifi County Integrated Development Plan (CIDP), the county has a monetary poverty rate of approximately 46.1%. This economic deprivation perpetuates a cycle of low resilience: households with minimal disposable income are unable to invest in livelihood improvement, asset replacement, or climate-adaptation measures. Consequently, each shock whether climatic, health-related, or economic deepens vulnerability, forcing negative coping strategies such as reduced meals, sale of productive assets, or child labour.

3.5.2 Vulnerabilities of Livelihood Options

All livelihood options in the ward while essential for household sustenance, expose the community to varying degrees of risk. Sand harvesting although the profitable activity, is environmentally and climatically vulnerable. Excessive extraction along seasonal rivers and near forested catchments disrupts groundwater recharge and accelerates soil erosion. During heavy rainfall or flooding, operations are halted, while prolonged droughts reduce sand replenishment. If surrounding ecosystems such as Mwangea Hill or Arabuko-Sokoke's drainage systems are degraded, sedimentation dynamics change, reducing the sustainability of sand deposits and increasing community vulnerability to water scarcity. Moreover, the activity itself

is not environmentally sustainable, as it alters river morphology, depletes non-renewable sand reserves, and destroys habitats critical for maintaining hydrological balance and biodiversity.

Charcoal burning depends heavily on forest resources from areas near Arabuko-Sokoke and local kaya forests. Degradation or protection restrictions within these ecosystems directly affect availability of fuelwood, forcing producers to travel longer distances or operate illegally. The practice contributes to deforestation, heightening exposure to climate shocks such as rising temperatures, erratic rainfall, and reduced soil fertility all of which undermine long-term livelihood security.

Motorbike operations provide some of the high monthly income, yet are highly sensitive to external shocks. Fluctuations in fuel prices, maintenance costs, and road conditions particularly during heavy rains and floods can significantly reduce income. The risk of accidents, lack of insurance, and vehicle breakdowns expose operators to catastrophic losses. This makes motorbike transport a high-income but high-risk livelihood.

Agriculture is highly sensitive to multiple climate and ecological shocks. Rain-fed crops face recurrent droughts and erratic rainfall, leading to low yields, while flooding damages fields and reduces harvests. Farmers also contend with pests, crop diseases, and human–wildlife conflicts, including crop raiding by monkeys, baboons, and elephants, which reduce productivity and income. These combined stresses force households to rely on purchased food and constrain investments in improved seeds, fertilizers, and irrigation, lowering resilience to future shocks.

Income from livestock production is highly vulnerable to drought, disease outbreaks, and water scarcity. Reduced pasture availability during dry spells and flooding leads to livestock mortality and lower productivity. Livestock grazing near forested areas, such as Arabuko Sokoke, are also exposed to human–wildlife conflict, with predation by wild animals and crop raids affecting both animals and feed availability. Disease prevalence often increases following extreme weather events, compounding economic losses and reducing household resilience.

The community relies on the sale of goods and services to supplement household income, but their livelihoods are frequently disrupted by environmental and climate-related challenges. Flooding and poor drainage can prevent access to markets and damage stock, while seasonal droughts and extreme weather reduce local incomes and purchasing power, lowering demand for traded goods. These conditions make small-scale trade an unpredictable and fragile source of livelihood.

Women and youth collect herbs from forests and sacred sites such as Arabuko Sokoke, Mwangea Hills Kaya Godoma, and Kaya Kadzitsoni, making this livelihood heavily dependent on forest health and seasonal resource availability. Overharvesting, forest degradation, and climate shocks such as drought and irregular rainfall reduce both the quantity and quality of herbs, while extreme heat can spoil perishable products. Balancing herb collection with household chores and caregiving further limits women's adaptive capacity, making income from this activity highly vulnerable to ecological and climatic stressors.

Households engage in firewood collection, water vending, and casual labour as supplementary sources of income, but all three livelihoods are highly sensitive to environmental and climate-related stresses. Firewood availability declines during droughts, while floods and heavy rains disrupt collection and transport, and overharvesting contributes to forest degradation. Water vending depends on reliable water sources, which are reduced during dry spells, and flooding can damage infrastructure or contaminate supplies. Casual labour, including farm or construction work, is irregular and often seasonal, with opportunities shrinking during extreme

weather events such as droughts or floods. The combination of low and unpredictable income from these activities, coupled with environmental pressures, makes households particularly vulnerable to shocks and limits their capacity to invest in more sustainable livelihoods.

3.5.3 Overall Classification and Implications

The evidence indicates that Sokoke Ward is highly exposed and has low-to-moderate resilience, largely because livelihoods depend on climate-sensitive natural resources. Households rely heavily on informal and resource-dependent activities, while high poverty levels and gendered unpaid labour further constrain adaptive capacity, increasing vulnerability to shocks. Enhancing resilience will require integrated measures that stabilize incomes in high-risk sectors, diversify livelihoods into higher-value and climate-resilient activities, reduce women's unpaid workload through labour-saving interventions, and strengthen social safety nets. In addition, improving infrastructure such as access roads, alongside nature-based solutions including sustainable management and restoration of terrestrial forests in Arabuko Sokoke, Mwangea Hills, Kaya Godoma, and Kaya Kadzitsoni can help the community move from reactive coping strategies towards proactive adaptation.

3.6 Analyzing causes of vulnerability.

3.6.1 Classification of Severity of the Vulnerabilities and Their Causes

The ward exhibits varying degrees of vulnerability across its four sampled villages, Ndugumnani, Mwele, Dzikudze, and Midugani, influenced by ecological setting, livelihood dependence, and adaptive capacity. Mwele and Ndugumnani are highly vulnerable due to their reliance on resource-dependent livelihoods that are sensitive to climatic shocks and ecological pressures. In contrast, Dzikudze and Midugani are moderately vulnerable, as households benefit from more diversified income sources that generate higher returns, though these livelihoods remain exposed to market fluctuations and climate-related risks.

Vulnerability is compounded by limited livelihood diversification, chronic poverty, and gendered unpaid labour, which reduces women's capacity to respond to shocks. Resource-dependent livelihoods in the ward are highly exposed to climatic and ecological stresses, while socio-economic constraints limit adaptive capacity. This classification highlights the multidimensional nature of vulnerability, where ecological, economic, and gender factors intersect to determine overall risk.

3.6.2 Identification of Unsafe Conditions

Unsafe conditions in Sokoke Ward refer to settings and circumstances that heighten exposure to climate-related hazards and undermine the safety of livelihoods and households. Settlements adjacent to Arabuko Sokoke Forest, such as Mwele, Ndugumnani and Dzikunze, face risks human–wildlife conflicts, which can damage crops, livestock, and forest-dependent resources. Resource depletion from activities such as herb collection, sand harvesting, and firewood gathering further exacerbates ecological stress, reducing the availability of critical livelihood resources and increasing exposure to hazards. In addition, villages experience infrastructure-related vulnerabilities, including poor road access, limited drainage, and inadequate market and storage facilities, which disrupt livelihoods during extreme weather events and limit rapid recovery.

From a gender and social perspective, unsafe conditions are intensified for women, who often travel long distances to collect water, firewood, and herbs, navigating degraded landscapes that increase physical risk. Limited access to resilient infrastructure such as raised storage facilities, safe processing areas, weather-resistant shelters, and early-warning or emergency services further heightens vulnerability for all households, particularly those headed by widows, Persons Abled Differently, and elderly.

3.6.3 Identification of Dynamic Pressures or Determinants of Vulnerability

Dynamic pressures represent the intermediate processes through which underlying socio-economic and institutional weaknesses are translated into direct, unsafe conditions that heighten vulnerability. In Sokoke Ward, these pressures manifest through rapid population growth, unregulated land-use change, overexploitation of natural resources, and weak enforcement of environmental by-laws, all of which interact to degrade ecosystems and undermine resilience.

Mwangea Hills, one of the ward's key ecological assets, extending across Mwahera, Dzikunze, and Ndugumunani sub-locations, is identified as a severely degraded area. Despite its recognition as a critical water tower, the hill has undergone extensive deforestation, with much of the original forest cover cleared and converted into farmland. The remaining vegetation consists mainly of scattered shrubs and degraded patches, leading to its classification as a fully degraded landscape. This degradation is primarily attributed to prolonged droughts that compelled local communities to clear forested areas for agriculture, further exacerbated by the lack of active restoration or protection initiatives. Current rehabilitation efforts are minimal and limited to small-scale on-farm forestry activities on selected plots.

Beyond the forest ecosystem of Mwangea Hills, increasing settlement expansion into Arabuko Sokoke, Kaya Godoma, and Kaya Kadzitsoni, driven by the scarcity of arable land has intensified competition for space and accelerated the loss of protective vegetation. Declining availability of forest resources has prompted households to adopt unsustainable coping strategies, including charcoal production, firewood collection, and overharvesting of herbs. While these activities provide short-term income, they further degrade the ecosystem, creating a feedback loop between livelihood pressures and environmental decline.

From an institutional perspective, weak coordination between county departments, local community structures, and national agencies constrains effective resource governance. Enforcement of conservation by-laws remains sporadic, while early warning systems for floods and droughts are either absent or poorly operationalized, reducing preparedness and timely response. Economically, fluctuating market prices for key forest-based resources such as firewood, charcoal, and herbs, combined with limited access to credit, continue to undermine household income stability. On the social front, the exclusion of women from decision-making forums, particularly in forest management committees, limits the inclusivity and sustainability of adaptation initiatives.

3.6.4 Identification of Underlying Causes of Vulnerability

The underlying causes of vulnerability in Sokoke Ward stem from interlinked socio-economic, environmental, and institutional deficiencies that collectively erode household resilience and adaptive capacity. Structural poverty remains pervasive, limiting access to education, technology, financial services, and diversified livelihood opportunities. According to the Participatory Climate Vulnerability Assessment (PCVA, 2025), the average household size in Sokoke Ward is 9 persons per household, indicating a high dependency ratio and increased pressure on limited natural and financial resources. Larger households require greater quantities of food, water, and energy, yet often rely on subsistence livelihoods and ecosystem-dependent activities such as sand harvesting, charcoal burning and small-scale agriculture. This dependency entrenches poverty cycles and increases exposure to environmental and economic shocks.

Environmental degradation constitutes another major underlying factor, manifesting in degradation of adjacent terrestrial forests such as Arabuko Sokoke and Mwangea Hills ecosystems. The loss of vegetative cover reduces the capacity of ecosystems to buffer against

climate-induced hazards, particularly floods, droughts, and strong winds, thereby increasing community exposure and reducing long-term ecological stability.

Institutionally, weak governance frameworks and limited coordination between the County Government of Kilifi, the Ward Climate Change Planning Committee (WCCPC), and relevant national agencies hinder the implementation of adaptive interventions. Resource constraints, overlapping mandates, and inadequate policy enforcement reduce the effectiveness of climate action at both ward and county levels. Furthermore, the absence of comprehensive early warning and information dissemination systems results in delayed community responses to climatic hazards, shifting the focus from preparedness to crisis management.

Gender inequality further compounds vulnerability. Patriarchal norms and customary land tenure systems restrict women's ownership and control of productive assets, constraining their participation in decision-making and their capacity to adopt climate-smart technologies or access credit facilities. Consequently, women remain overrepresented in low-return and climate-sensitive livelihood activities, limiting their economic independence and adaptive potential.

3.6.5 Prioritization of Causes

Prioritization of the identified causes was achieved through participatory ranking during the PCVA exercises, considering frequency, severity, and reversibility of impacts. The highest priority causes were poverty, environmental degradation, and weak institutional support systems. Poverty restricts adaptive investments and perpetuates reliance on climate-sensitive livelihoods, while environmental degradation reduces the ecosystem's regenerative capacity. Weak institutional mechanisms delay response and hinder the implementation of ward-level adaptation initiatives.

Medium-priority causes include gender inequality, population pressure, and lack of access to climate information, all of which significantly influence vulnerability dynamics. Addressing these priority causes requires an integrated approach, strengthening governance structures, enhancing livelihood diversification, and promoting gender-responsive adaptation strategies. Interventions such as terrestrial and Kaya forests restoration, provision of microcredit for women's enterprises, and establishment of community-based early warning systems are essential to reduce long-term vulnerability and build resilience across all villages in the ward.

4 Chapter 4: Gender/Age Vulnerability Analysis for Climate Hazards

Gender and age critically determine exposure, sensitivity, and adaptive capacity in Sokoke Ward, where the main climate hazards; drought, floods, pests and diseases, human–wildlife conflicts from the Arabuko-Sokoke Forest, environmental degradation driven by charcoal burning, and whirlwinds/strong winds, interact with deep-rooted socio-cultural inequalities and chronic poverty (UN Women, 2022; IPCC, 2022). These stressors disproportionately affect women, girls, youth, elderly persons, and Persons Abled Differently (PADs), whose vulnerability is shaped by limited access to productive assets, restricted mobility, gendered labour divisions, and unequal decision-making power at household and community levels (Table 1).

A central vulnerability driver among women and girls in Sokoke is the heavy and undervalued burden of unpaid domestic and care work, including childcare, cooking, water and firewood collection, elder care, cleaning, and small-scale farming. During PCVA exercises, this workload was valued at KES 1,000 per day, yet remains invisible within formal economic accounting (KNBS, 2023; UNDP, 2022). Climate hazards intensify this workload by increasing the time required to secure basic needs, thereby deepening "time poverty" and limiting women's participation in livelihood diversification, leadership spaces, and climate adaptation activities (ILO, 2018; UN Women, 2023).

Intra-household power imbalances in Sokoke further limit women's adaptive capacity. Many women lack control over land, farming decisions, livestock use, income allocation, and children's schooling. This undermines their ability to adopt climate-resilient technologies or engage effectively in local adaptation processes. Climate-related stress often heightens household tensions and contributes to gender-based violence (GBV), including emotional, physical, and economic abuse, which restrict women's mobility, erode confidence, and weaken their participation in social and economic spheres. Rising cases of femicide nationally further deepen fear and constrain women's agency, reducing adaptive capacity (UNFPA, 2023; Kenya Police Service, 2024).

Drought is one of the most significant hazards in Sokoke Ward, contributing to water scarcity, livestock deaths, crop failure, and household food insecurity. Women and girls walk long distances, often up to 5 km, to fetch water and firewood, exposing them to heat stress, dehydration, musculoskeletal strain, and protection risks such as harassment (UNICEF, 2022). Female-headed households are highly affected due to limited labour support and reduced financial buffers. Men experience reduced productivity in farming and livestock rearing, increasing psychological stress and constraining household income. Youth (18–35) face declining opportunities for casual labour and apprenticeships, increasing reliance on unsafe coping strategies (UNFPA, 2022). Elderly individuals and PADs struggle to access food, water, and relief support due to mobility challenges (WHO, 2021).

Flooding in Sokoke disrupts transportation, destroys homes, damages small businesses, and affects feeder roads and local markets. Women are disproportionately affected as they care for sick or displaced family members, manage contaminated water, salvage household assets, and absorb the increased domestic labour burden. Flood-induced market closures reduce women's income from informal businesses. Men lose income from casual labour and on-farm activities due to destruction of agricultural fields. Youth face disruptions in schooling and income-generating activities, heightening vulnerability to exploitation and unsafe behaviour (UNESCO, 2021). Elderly persons and PADs experience substantial challenges evacuating, heightening exposure to waterborne diseases, injury, and neglect in overcrowded shelters (WHO, 2020).

Pests and livestock/crop diseases threaten agricultural productivity in Sokoke Ward, which is heavily dependent on maize, cassava, coconut, and poultry production. Women, who

dominate subsistence farming and small livestock management, experience reduced yields, post-harvest losses, and declining food security. Their adaptive capacity is constrained by limited access to pest-control inputs, certified seeds, veterinary services, and agricultural extension (FAO, 2021). Men face reduced income from compromised farms and diseased livestock. Youth lose wage labour opportunities, especially in planting and harvesting seasons. Elderly persons and PADs experience heightened food insecurity due to dependence on household food stocks.

Human–wildlife conflict is a major hazard in Sokoke Ward, particularly for households bordering the Arabuko–Sokoke Forest, where wildlife such as elephants, baboons, buffaloes, bushpigs, and monkeys invade farms, destroy crops, damage property, and threaten safety. Women and girls limit movements to avoid wildlife encounters, reducing access to farms, water points, firewood areas, and markets. Limited documentation and restricted decision-making power hinder their access to compensation mechanisms. Men face increased risk guarding farms, often at night, resulting in injury risks and recurrent crop losses. Youth helping with farm protection face physical danger and disruptions in schooling. Elderly persons and PADs are extremely vulnerable due to slow mobility and inability to escape during wildlife encounters, increasing emotional stress and dependency (UNEP, 2022).

Sokoke Ward has experienced extensive environmental degradation due to illegal charcoal production and tree felling, particularly around forest edges and communal lands. This deforestation reduces vegetation cover, accelerates soil erosion, disrupts microclimates, and diminishes access to firewood and non-timber forest products. Women must walk longer distances to obtain fuelwood, increasing exhaustion and reducing time for income-generating activities. Men experience reduced farm productivity due to soil degradation and decreased water retention in soils. Youth lose opportunities in nature-based activities such as beekeeping, seedling propagation, and eco-tourism enterprises. Elderly persons and PADs face heightened dependency due to strained access to natural resources (UNDP, 2022).

Whirlwinds and strong winds are increasingly common in Sokoke due to deforestation and changing local climatic patterns. These events damage homes, uproot trees, destroy crops, and disrupt schooling. Women face increased caregiving responsibilities and must secure household property during storms. Men experience destruction of farm structures, livestock sheds, and stored produce. Youth face injury risks from collapsing structures and flying debris, while elderly persons and PADs have difficulty reaching safe shelter, increasing their exposure to danger (WHO, 2020)

Gender-based violence (GBV) further compounds these gendered vulnerabilities, functioning both as a climate-induced risk and as a structural barrier to resilience in Pongwe Kikoneni Ward. Emerging global evidence demonstrates a strong association between climate stressors and rising GBV incidence, with every 1°C increase in temperature linked to a 4.7% rise in intimate partner violence (UN Spotlight Initiative, 2024). Heatwaves have been associated with a 28% increase in femicide, while climate-driven displacement heightens risks of child marriage, sexual exploitation, and trafficking (UN Spotlight Initiative, 2024). In Kenya, rising temperatures and socioeconomic stressors have coincided with escalating femicides with over 930 cases documented between 2016 and 2024, highlighting how climate stress intersects with entrenched gender inequalities (Kenya Femicide Data Study, 2024). These dynamics are most acute in fragile, resource-dependent communities such as Pongwe Kikoneni, where drought, floods, and livelihood loss amplify household tensions, reduce economic security, and disproportionately expose women and girls to physical, psychological, and sexual violence. GBV undermines climate resilience by restricting women's mobility, decision-making autonomy, access to productive assets, and participation in ward-level adaptation processes (UN Women, 2022). As such, GBV must be integrated into anticipatory climate action, beyond response and

preparedness, through early warning systems, forecast-based triggers, community surveillance, and pre-positioned support services coordinated with mandated county structures such as the County Gender Department, Social Services, child protection units, and GBV coordination mechanisms that programmes like COSME have actively engaged. Embedding GBV mitigation within resilience planning ensures that climate adaptation measures safeguard rights, safety, and justice, which are foundational to sustainable and inclusive resilience building

Table 3: Gender/Age Vulnerability Analysis for Climate Hazards

Women	Men	Youth	Elderly	PADs
Drought				
• Female-headed households bear highest burden of water, food, and fuelwood provision • Walk longer distances to depleted water sources → fatigue, dehydration, back strain • Reduced time for childcare, rest, and income generation • Increased psychological stress due to food shortages	• Livelihood insecurity from reduced crop yields, livestock productivity, and forest resources • Increased financial pressure and stress • Reduced availability of agricultural labour opportunities	• Reduced productivity and income from casual labour losses • High idleness → increased risk of crime, drugs, and unsafe coping strategies • Higher incidence of early marriages and pregnancies among girls	• Mobility limitations restrict access to water and food • Increased susceptibility to heat stress, dehydration, and hunger • Dependence on caregivers worsens during drought periods	• High dependency for water, food, and healthcare access • Mobility constraints limit ability to queue for relief supplies • Increased risk of neglect during prolonged shortages
Floods				
• Increased caregiving burdens for sick or displaced family members • Loss of income from informal trade due to market inaccessibility	• Loss of fishing, transport, and casual labour opportunities • Psychological stress from inability to provide for families • Damage to tools and assets disrupts livelihoods	• School closures disrupt learning and training • Loss of casual labour -increased idleness, crime, and drug use • Girls particularly vulnerable to early	• High susceptibility to waterborne diseases and cold exposure • Limited mobility restricts evacuation • Greater dependence on others for shelter, food, and medicine	• Severe mobility barriers during evacuation and accessing aid • High risk of injury and neglect in shelters • Limited access to emergency services

Women	Men	Youth	Elderly	PADs
• Contaminated wells force longer distances to find safe water • Elevated psychosocial stress		marriage and pregnancy		
Pests and diseases				
• High exposure due to their role in managing subsistence farms, and small livestock • Reduced food security and income from harvest losses • Limited access to pest-control technologies and extension services	• Income loss from crop and livestock declines • Increased stress due to reduced labour demand and food shortages • Higher pressure to diversify livelihoods	• Loss of agricultural casual labour opportunities • Increased unemployment and financial vulnerability • Greater exposure to negative coping behaviours	• Increased dependence for food and care during outbreaks • Reduced capacity to protect crops and livestock • Nutritional deficits worsen chronic illnesses	• High dependency on family for food and adaptive support • Limited access to adapted technologies and extension services • Increased vulnerability from household food shortages
Human wildlife conflicts				
• Restricted mobility due to fear of wildlife attacks • Loss of crops and livestock increases household food insecurity	• High risk of injury when guarding farms and livestock, often at night • Economic strain from repeated crop and livestock losses	• Increased risk when helping to guard farms • Reduced productivity and fewer livelihood opportunities • Higher vulnerability to	• Very limited capacity to respond to wildlife encounters • Increased dependence after livestock or crop losses	• Inability to flee or protect themselves during wildlife incidents • Extremely high dependence on caregivers for safety and recovery

Women	Men	Youth	Elderly	PADs
• Limited access to compensation processes	• Pressure to seek alternative income sources	migration and exploitation	• Heightened food insecurity	• High vulnerability during compensation or reporting processes
Environmental Degradation				
• Reduced access to firewood, herbs, and NTFPs • Walk longer distances (5+ km) for resources leading to exhaustion • Less time for childcare and economic activities	• Declining fish stocks increase offshore travel distance and fuel costs • Higher fishing risks • Reduced forest resource availability increases livelihood strain	• Fewer forest-based livelihood options (beekeeping, nurseries, eco-tourism) • Increased involvement in illegal extraction when opportunities shrink • Rising idleness and economic vulnerability	• Mobility challenges increase dependency during resource scarcity • Physical strain from long distances to collect resources • Heightened vulnerability to malnutrition	• Limited access to degraded resource areas • Exclusion from resource-based livelihood opportunities • Increased reliance on caregivers
Strong winds/Whirl winds				
• Increased caregiving demand after home damage • Higher risk securing household assets during storms	• High danger at sea and gear damage • Increased risk of capsizing during strong winds	• High injury risk from collapsing structures or debris • Limited ability to seek safe shelter	• Difficulty accessing safe shelter quickly • High injury and exposure risk	• Severe challenges evacuating safely • High exposure to falling debris and structural collapse

4.1 Children's Vulnerability to Climate Hazards and Key Mitigation Measures in Sokoke Ward

Children in Sokoke Ward are among the most climate-vulnerable groups due to their dependence on caregivers, rapid physiological development, and limited ability to anticipate or respond to climate shocks (UNICEF, 2022). Their vulnerability is heightened by poverty, food insecurity, limited WASH access, and frequent disruptions to schooling, all of which are intensified by the ward's major hazards; drought, floods, pests and diseases, human–wildlife conflicts from Arabuko-Sokoke Forest, environmental degradation from charcoal burning, and whirlwinds/strong winds.

Drought exposes children to acute malnutrition, dehydration, anaemia, and stunting as households rely on unsafe water sources and reduced farm production (WHO, 2020). Many children, especially girls, are tasked with water collection, forcing them to walk long distances under high temperatures, which generates heat stress, fatigue, and protection risks such as harassment (UNICEF, 2022). These responsibilities contribute to school absenteeism, poor concentration, and limited participation in early learning (UNESCO, 2021).

Floods in Sokoke destroy homes, disrupt WASH infrastructure, and damage schools, increasing children's exposure to drowning, diarrhoeal diseases, cholera, skin infections, and respiratory illnesses (WHO, 2020). When schools serve as evacuation centres, learning continuity is interrupted and school feeding programmes, critical for many children, are disrupted (UNICEF, 2021). The instability associated with displacement causes psychosocial impacts such as anxiety, depression, nightmares, and behavioural regressions (UNFPA, 2021).

Pests and diseases undermine food production and livestock health, reducing dietary diversity for children. Household food insecurity leads to reduced meal frequency, micronutrient deficiencies, and delayed physical and cognitive development (FAO, 2021). Children in food-insecure households tend to miss school more often and struggle academically due to poor nutrition (UNDP, 2022).

Human–wildlife conflicts (HWCs) from the Arabuko–Sokoke Forest expose children to direct physical danger during routine daily activities such as grazing livestock, collecting firewood, or accompanying caregivers to the farm. Encounters with elephants, baboons, monkeys, and bushpigs create emotional trauma, panic attacks, and long-term fear responses in children (UNEP, 2022). Families often restrict children's movement during peak wildlife activity, which limits access to school, play spaces, and peer interactions essential for psychosocial development.

Environmental degradation from charcoal burning reduces vegetation cover, increases dust and smoke levels, and diminishes access to clean water and fuelwood. Children exposed to charcoal smoke are at increased risk of respiratory infections, persistent coughs, asthma, and eye irritation (WHO, 2021). As households travel further to collect biomass, children—especially girls—bear increased workloads, reducing study time and rest (UNDP, 2022).

Whirlwinds and strong winds pose direct physical risks, as children are more vulnerable to injury from collapsing roofs, falling branches, and debris. School infrastructure is frequently affected, leading to class disruptions, trauma, and emotional insecurity (UNESCO, 2021).

In Sokoke Ward, climate-related stress often forces households into harmful coping strategies, including early marriage, withdrawal of children from school, child labour, and transactional interactions among adolescents, especially during prolonged drought or after severe flood events (UNFPA, 2022). These practices undermine children's rights and perpetuate intergenerational poverty.

Protecting children requires multi-sectoral, child-sensitive climate adaptation, including strengthening community and school-based early warning systems (UNDRR, 2020), enhancing nutrition surveillance and emergency feeding programmes, and implementing climate-resilient WASH systems such as protected boreholes, elevated latrines, and rainwater harvesting (UNICEF, 2022; WHO, 2020). Climate-proofing educational infrastructure with storm-resistant roofing, reinforced classrooms, and shaded playgrounds is essential for safeguarding learning continuity (UNESCO, 2021). Provision of mental health and psychosocial support, child-friendly spaces, trauma-informed counselling, and family-tracing services further protects children during displacement or recovery (UNFPA, 2021). Strengthening social protection, including school feeding, bursaries, and child grants, helps households avoid negative coping strategies and supports children's well-being during climate shocks (UNDP, 2022).

5 Chapter 4: Adaptation Actions

5.1 Introduction

Building on the findings of the hazard and vulnerability analysis, this chapter outlines the adaptation actions designed to strengthen climate resilience in Sokoke Ward. These actions were developed through participatory consultations that utilized Participatory Climate Vulnerability Assessment (PCVA) tools and impact chain analysis to ensure that proposed measures align with the ward's ecological characteristics and socio-economic context. The interventions are tailored to address both the immediate climatic stresses, such as droughts, floods, Pests and diseases, environmental degradation, human wildlife conflicts and Whirl winds, and the underlying structural vulnerabilities related to poverty, ecosystem degradation, and institutional capacity gaps.

The proposed measures encompass short-, medium-, and long-term strategies. Short-term actions include community awareness creation, institutional strengthening, and capacity-building initiatives. Medium- and long-term interventions involve climate-resilient infrastructure development, ecosystem restoration, and the promotion of nature-based and technological solutions. For instance, the installation of water harvesting and storage systems is intended to mitigate drought, while drainage restoration will reduce the impacts of flooding.

The formulation of these interventions integrates local knowledge with scientifically validated best practices drawn from Kenya's National Adaptation Plan (2015–2030), the National Climate Change Action Plan (2023–2027), and the Kilifi County Climate Change Action Plan (2023–2027). Gender and social inclusion are central to these actions, ensuring that women, youth, and persons with disabilities (PADs) are actively involved in planning and benefit equitably from all adaptation measures. Women, who often shoulder greater burdens during droughts and resource scarcity, are supported through programs in livelihood diversification, leadership, and financial empowerment, while youth are engaged in technology-driven adaptation initiatives such as climate information sharing and ecosystem monitoring. Implementation will occur across multiple levels, from households and community groups such as CFAs and WRUAs to ward, county, and national institutions, under the coordination of the WCCPC.

Currently, there are ongoing community-led projects in Sokoke Ward that complement these adaptation priorities, including tree planting initiatives and local savings and credit groups (VSLAs) that support climate-resilient livelihoods. These initiatives demonstrate the community's commitment to sustainable adaptation and provide a strong foundation upon which the proposed actions will build. Implementation of the adaptation measures will occur at multiple levels i.e., household, community, ward, county, and national, under the coordination of the Ward Climate Change Planning Committee (WCCPC). Collaborative engagement among local organizations, BMUs, WRUAs, and relevant government institutions will be essential to ensure coherent, inclusive, and sustainable delivery of the adaptation agenda across Sokoke Ward.

5.2 Existing climate-resilient and livelihood-enhancing projects

Sokoke Ward currently benefits from community-led initiatives, institutional partners, and development organizations implementing climate-resilient and livelihood-enhancing projects (Table 4). These interventions collectively strengthen local adaptive capacity, promote ecosystem restoration, and support sustainable resource use. Existing strategies focus on forest restoration and community-based financial inclusion, demonstrating strong collaboration between the community, civil society, and county agencies.

Key ongoing initiatives include the implementation of conservation and livelihood resilience programs led by Plan International, World Vision Kenya, Youth in Sustainable Agriculture (YISA), and One Acre Fund. These initiatives collectively strengthen climate resilience, restore degraded ecosystems, and enhance household adaptive capacity through integrated social and environmental approaches.

Plan International continues to champion the Conservation and Sustainable Management of Coastal and Marine Ecosystems (COSME) program, which focuses on community empowerment through Village Savings and Loan Associations (VSLAs), table banking, and gender justice initiatives. Through these efforts, women's economic participation has increased, strengthening household income diversification and reducing vulnerability to climate shocks.

World Vision Kenya is implementing the Restore Africa Project, which promotes tree planting, forest landscape restoration, and soil conservation. These interventions directly contribute to the rehabilitation of degraded areas such as Mwangea Hills and support the restoration of culturally and ecologically significant sites, including sections adjacent to Arabuko Sokoke Forest and the sacred Kaya forests.

Youth in Sustainable Agriculture (YISA) is supporting youth-led enterprise development by providing funds and technical assistance to establish small-scale businesses. This initiative has reduced dependency on climate-sensitive activities such as rain-fed farming and charcoal production, encouraging a transition towards sustainable and diversified livelihoods.

One Acre Fund, through the Tupande Project, supplies farmers with high-quality seedlings, farm input packages, and agronomic training to enhance agricultural productivity and resilience. By improving access to inputs, markets, and extension support, the program contributes to food security and improved household incomes.

Collectively, these interventions complement ongoing county strategies, including the Kilifi County Climate Change Action Plan (2023–2027) and the County Integrated Development Plan (CIDP 2023–2027), by promoting community-based adaptation, gender inclusion, and ecological restoration. They also reinforce national frameworks on climate resilience, sustainable agriculture, and forest landscape restoration.

Table 4: Existing climate-resilient and livelihood-enhancing projects in Sokoke Ward

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
Capacity Building on Forestry, VSLA and Gender Empowerment Initiatives	Plan International	Mwele, Dzikudze, Midugani, Ndugumnani	Social Protection / Livelihood Diversification / Gender Empowerment	Promotes women's economic participation through VSLAs and table banking; strengthens household resilience and aligns with the Kilifi County Climate Change Action Plan (2023–2027) and Gender Mainstreaming Strategy.
Restore Africa Project (Forest and Landscape Restoration)	World Vision Kenya	Mwele, Dzikudze, Midugani, Ndugumnani	Environmental Restoration / Forest Landscape Rehabilitation	Supports reforestation and soil conservation in degraded ecosystems
Youth Enterprise Support and Climate-Resilient Livelihoods	Youth in Sustainable Agriculture (YISA)	Mwele, Dzikudze, Midugani, Ndugumnani	Economic Empowerment / Livelihood Diversification	Provides youth with funding and technical training to start small-scale enterprises, reducing reliance on climate-sensitive livelihoods
Tupande Project (Climate-Smart Agriculture and Farmer Support)	One Acre Fund	Mwele, Dzikudze, Midugani, Ndugumnani	Sustainable Agriculture / Food Security	Enhances agricultural productivity and resilience through high-quality seedlings, inputs, and training; aligns with the National Climate-Smart Agriculture Strategy (2017–2026) and County Food Security Framework

5.3 Adaptation actions

Adaptation actions in Sokoke Ward are designed as an integrated and gender-responsive climate resilience framework that protects livelihoods, strengthens ecosystems, and safeguards the well-being of all population groups across the four villages; Dzikudze, Midugani, Mwele and Ndungumnani (Table 5). These actions respond to six priority climate hazards; drought, floods, pests and diseases, strong winds, environmental degradation and human–wildlife conflicts and are anchored in the Kenya Climate Change Act (2023), the National Adaptation Plan (2015–2030), the Kilifi County Climate Change Action Plan (2023–2027), and national frameworks on gender equality, child protection, and sexual and reproductive health and rights (SRHR). Implementation follows a phased approach comprising short-term interventions (≤1 year), medium-term measures (≤3 years), and long-term interventions (>3 years), all anchored in gender equity, social inclusion, SRHR continuity, and community ownership to ensure climate benefits are distributed fairly and sustainably.

Under Thematic Area One, Water Security and Management focuses on strengthening household, institutional and community access to reliable and safe water resources across Dzikudze, Midugani, Mwele and Ndungumnani villages. In the short term, households will prioritize rainwater harvesting through the installation of gutters and storage tanks at an estimated cost of KES 50,000 per household, reducing the time burden on women and girls who undertake most water collection. Immediate ecosystem rehabilitation will be undertaken through maintenance of the existing well and borehole in Midugani village, budgeted at KES 500,000, thereby restoring water storage capacity for domestic and livestock use. These short-term actions are complemented by sensitization of women and youth on household water treatment practices, implemented through barazas facilitated by CHPs, NGOs and KIMAWASCO at KES 20,000 per village.

Medium-term measures focus on institutional strengthening through the installation of 10 m^3 high-capacity water tanks and rooftop harvesting systems in one school per village, except in Midugani where a church will serve as the anchoring institution, costed at KES 400,000 per institution. These facilities will support school WASH systems and provide fallback water supplies during drought. The programme will also provide solar powered water pumps on existing wells in Midugani village budgeted at KES 70,000, to ensure perennial water availability for domestic and small-scale irrigation use. These interventions will be delivered collaboratively through Kilifi County Government (FLLoCA), NGOs, private sector partners and KWAWASCO.

Long-term actions advance drilling and solarization of one borehole per village, each budgeted at KES 5,000,000 to enhance water reliability for households and women-led livelihood systems. Each solar pumping unit is estimated at KES 70,000. To further enhance resilience, the construction of water pans in Dzikunze, Midugani, Mwele and Ndugumnani at KES 3,000,000 per water pan will store excess rainfall and provide year-round supply for irrigation and livestock. These long-term interventions collectively aim to reduce climate-related water stress while improving gender-equitable access to safe water.

Thematic Area Two focuses on food security, biodiversity, ecosystem restoration and livelihoods through climate-smart agriculture and resilience design from homestead to watershed level. In the short term, women and youth will be capacitated on resilience design techniques, soil fertility enhancement, water retention using zai pits, mulching and efficient farm layout, delivered through county extension services and NGOs at KES 10,000 per farmer per 12-month cycle. Training will also cover climate-smart agriculture, including drought-tolerant crop varieties (cassava, katumani maize, sorghum), hay and silage making, as well as sensitization on index-based crop and livestock insurance.

Medium-term interventions include distributing certified seeds, fertilizers, pesticides and livestock inputs to trained groups in all four villages. Production support is costed at KES 55,000 per acre for maize, KES 40,000 per acre for cassava, and KES 10,000 per livestock farmer. Select women- and youth-led farms will be supported to fully implement resilient design techniques, forming demonstration farms for continuous peer learning. Establishing a Ward-level Farmer Field School (FFS) will institutionalize resilient farming practices and ensure continuous technical support.

Long-term actions prioritize market strengthening by aggregating produce from women and youth to enhance bargaining power and value addition. Livestock diversification will be supported through pasture establishment at KES 55,000 per acre, creation of fodder seed banks, and distribution of Gala goats (KES 15,000 each) and dairy cows (KES 100,000 each). These interventions reduce climate sensitivity while improving livelihood stability.

In addition, the thematic area focuses on restoring degraded ecosystems and enhancing nature-based solutions. In the short term, women, youth and PADs will be trained to establish nurseries for priority tree species for agroforestry, boundary planting and woodlot establishment. Training is budgeted at KES 5,000 per trainee per day, and each nursery will cost KES 75,000 to establish. Training on briquette production will also be provided to reduce dependence on charcoal and mitigate forest degradation.

Medium-term interventions include providing households with 100 priority seedlings per home at KES 5,000 per household, rehabilitating degraded sections of Mwangea hills and Kaya Kadzistoni KES 150,000 per acre. Provision of energy-saving jikos (KES 3,000) and kilns (KES 7,000) will reduce pressure on forest resources, enhancing long-term sustainability.

Thematic Area Three strengthens adaptive capacity through diversified income streams and social safety nets. Short-term actions include training women, youth and PADs in alternative livelihoods such as kitchen gardening, beekeeping, poultry production and briquette making, delivered at KES 5,000 per trainee per day. VSLA governance structures will be strengthened to promote financial literacy and enhance adaptive capacity.

Medium-term actions include provision of livelihood inputs, seeds and organic fertilizers, modern beehives (KES 7,000) and egg incubators (KES 100,000), to facilitate enterprise establishment. Women will be supported to establish small scale enterprises (stalls) at household level across all the villages.

Thematic Area Four addresses structural vulnerabilities to floods, strong winds and drought. Short-term actions include training community emergency response teams across all four villages at KES 5,000 per trainee, alongside tree planting in Ndungumnani to serve as a windbreak system. These foundational measures strengthen preparedness and immediate response to climate hazards.

Medium-term interventions include developing and operationalization of Early Warning Systems (EWS) in all villages at KES 70,000 per village, improving access to predictive information on strong winds, droughts, and pest outbreaks.

Long-term actions involve equipping community emergency response teams (PPEs, first aid kits) in all villages and Construction a health facility in Mwele Village. All long-term infrastructural works will be guided by BOQs and implemented in partnership with Kilifi County Government, KMD, NDMA, Kenya Red Cross and NGOs.

Thematic Area Five addresses rising human–wildlife conflicts (HWC) especially in Mwalewa C and villages near Gonja Forest. Short-term actions include community sensitization on conflict hotspots, safe behaviour to reduce encounters, and reporting mechanisms to KWS through the emergency hotline. Each sensitization forum will cost KES 20,000 per village. Long-term strategies involve strengthening coordination between KWS, county authorities and communities to institutionalize conflict mitigation, improve surveillance and enhance access to compensation pathways.

Thematic Area Six centers on integrating gender equality, youth empowerment and inclusion of PADs into climate adaptation. In the short term, targeted GBV anticipatory actions will be developed for drought, floods and strong winds, ensuring safe access to services for women and girls. Existing women's associations such as VSLAs will be used to disseminate information on rights, helplines and GBV response systems at KES 30,000 per VSLA. Youth boot camps addressing drug abuse, reproductive health and life skills will be implemented at KES 100,000 per camp, and children and youth will be vaccinated at local health facilities to reduce outbreak risks during climate shocks.

Medium-term actions include assessing and refining Early Warning Systems for climate-linked GBV anticipatory responses, and strengthening inclusive governance through training women, youth and PADs at KES 5,000 per person per day. GBV response mechanisms, including transfer vehicles and specialized first-aid and rape kits, will be procured based on BOQs. School feeding and scholarship support will be anchored on needs assessments to safeguard climate-vulnerable learners.

Long-term interventions include developing and implementing a comprehensive Ward GBV anticipatory action plan (costed at KES 1,000,000 for consultancy services), constructing a county-level drug rehabilitation centre, and facilitating PADs to access mobility devices, assistive technology and entrepreneurial training through stakeholder engagement forums budgeted at KES 50,000.

Table 5: Sokoke Ward Adaptation Actions

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
Thematic area 1: Water Security and Management				
Rainwater harvesting through Installation of gutters on house roofs and water tanks at household level in Dzikunze, Midugani, Mwele and Ndugumnani villages	Rainwater harvesting through installation of gutters and high-capacity water tanks (10m^3) in schools– one school in each village to enhance water access by women and children In Midugani village installation to be done at EAPC church due to lack of a school	Drilling and solarizing boreholes in each villages-one in each village	Kilifi County Government-FLLoCA, NGOs and Private Sector	50,000/HH, 400,000/School, 5,000,000/borehole
Rehabilitation and maintenance of the existing well and borehole in Midugani village as well as the water pan in Dzikunze to provide water for women and youth for domestic needs	Strategic (where embankment for water intake is viable) installation of solar powered water pumps on existing wells to reduce women workload (turn-around time) in Midugani Village	Construction of water pans in Dzikunze, Midugani, Mwele and Ndugumnani villages to reduce water scarcity and workload among women and youth	KIMAWASCO, Kilifi County Government-FLLoCA, NGOs and Private Sector	70,000/ water pump, 50,000/well rehabilitation, 500,000/borehole rehabilitation, 1,000,000/water pan rehabilitation 3,000,000/water pan construction
Sensitizing women and youth on water treatment in Dzikunze, Midugani, Mwele and Ndugumnani villages			Of boreholes , Kilifi County Government, CHPs, NGOs and Private Sector	20,000/village baraza
Thematic area 2: Food security, ecosystem restoration and livelihoods				
Training and supporting women and youth on Resilience Design techniques from homestead to watershed level (Soil fertility improvement, water conservation/retention (e.g. Zai Pits), farm design to maximize on the time and use of local inputs	Supporting select women and youth lead farmers to fully implement resilient design techniques including climate smart agriculture to serve as demonstration farms for continuing learning by other farmers to enhance sustainability of the interventions	Establishing a Farmer Field School in the Ward	Kilifi County Government, KARLO, NGOs and Private Sector	Farmer Field Schools (FFFs) 10,000 per farmer for a 12-month cycle

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
Training women and youth on Climate Smart Agriculture (CSA) (drought tolerant crops such as katumani maize cassava, sunflower, groundnuts, cashewnuts, sorghum), silage/hay making to address livestock feed challenges during drought in Dzikunze, Midugani, Mwele and Ndugumnani villages	Provision of farm inputs (certified seed varieties for drought tolerant crops, organic and inorganic fertilizers, pesticides) and extension services across Dzikunze, Midugani, Mwele and Ndugumnani villages to trained women and youth groups	Aggregating women and youth farmers' produce to enhance farm produce marketability and enhance value addition and agribusiness opportunities Training women, youth and PADs on cassava value addition in Dzikunze, Midugani, Mwele and Ndugumnani villages		Production costs: Maize 55,000 per acre and Cassava 40,000 per acre 10,000 for livestock farmers per year or crop cycle (Free if it is provided by the government)
Sensitize and facilitate access to information on crop and livestock insurance (index-based drought coverage) targeting women and youth groups involved in CSA	Establish livestock fodder seed banks, pastures, and introduce drought-tolerant livestock breeds e.g., Gala goats/Dairy cows for women and youth groups in Dzikunze, Midugani, Mwele and Ndugumnani villages			Village barazas at 20,000, 55,000 per acre for pasture, 15,000 per Gala goat, 100,000 per dairy cow
Facilitate/ train women and youth to establish tree nurseries of priority species (Annex 1) for on-farm tree production (Agroforestry, woodlots and boundary planting) in Dzikunze, Midugani, Mwele and Ndugumnani villages	Rehabilitate degraded areas in Mwangea hills and Kaya Kadzistoni		Kilifi County Government, KFS, KEFRI, CFAs, NGOs, Private Sector	5,000 per day per trainee (40 pax from the five villages) 75,000 per nursery 150,000 per acre for rehabilitation
Training women, youth and PADs on making organic briquettes (coconut husks, cassava peels) in Dzikunze, Midugani, Mwele and Ndugumnani villages Sensitization on climate change and environmental	Provision of energy saving jikos and kiln across Dzikunze, Midugani, Mwele and Ndugumnani villages			5,000 per day per trainee (40 pax from the four villages) Village barazas at 20,000 3,000 energy saving jiko & 7,000 per Kiln

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
conservation in Dzikunze, Midugani, Mwele and Ndugumnani villages				
Thematic area 3: Diversified Livelihoods and Social Protection				
Training women, youth and PADs on alternative livelihoods such as beekeeping and poultry farming in Dzikunze, Mwele, Midugani and Ndugumnani villages	Provision of alternative livelihoods establishment materials/Equipment; modern beehives, incubators in Dzikunze, Mwele and Ndugumnani villages		Kilifi County Government, NGOs, Private Sector	5,000 per day per trainee (40 pax from the four villages) 7,000-Modern beehive, 100,000-eggs incubator
	Provision of relief food and water to households during drought periods in Dzikunze, Mwele and Ndugumnani villages Supporting women, youth and PADs in establishing small scale enterprises (stalls) at household in Dzikunze, Mwele and Ndugumnani villages		Kilifi County Government, NGOs, Private Sector	On a need basis 50,000/HH
Thematic area 4: Community infrastructure and disaster management				
Train selected community emergency response teams in Dzikunze, Midugani, Mwele and Ndugumnani villages Undertake Tree planting to address strong winds in Ndugumnani	Develop and operationalize Early Warning Systems (EWS) for drought, human-wildlife conflicts and strong winds (forecasts via chiefs, barazas, KWS and media) in Dzikunze, Midugani, Mwele and Ndugumnani villages	Equip community emergency response teams (PPEs, first aid kits) in Dzikunze, Midugani, Mwele and Ndugumnani villages Construct a health facility in Mwele Village	KMD, KWS, NDMA Kilifi County Government, Ministry of Interior and National Administration Kenya Red Cross, NGOs and Private Sector	5,000 per day per trainee 70,000 per village for EWS BOQ for equipment and health facility
Thematic area 5: Conflicts resolution (Human Wildlife Conflicts)				
Sensitize Dzikunze, Mwele, Midugani and Ndugumnani villages on potential conflict hotspots, minimizing human-	Promote chili cultivation (provide training and quality seeds) and apiculture (provide training and hives) along forest–farmland boundaries as		KWS, Kilifi County Government, NGOs, Private sector	Village barazas at 20,000 350/packet-chilli

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
wildlife interactions and reporting to KWS (hotline) as well as compensation procedures	a Nature-Based Solution (NBS) in Dzikunze, Midugani, Mwele and Ndugumnani villages			5,000,000- Death 3,000,000- Permanent Disability 2,000,000- Any other injury Crop/Property destruction- compensation based on market rates
Thematic area 6: Gender and social inclusion to enhance adaptation action				
Design targeted GBV anticipatory actions for drought and floods climate hazards in Dzikunze, Midugani, Mwele and Ndugumnani villages	Develop and implement a Ward GBV anticipatory action plan in collaboration with relevant partners	Establish GBV recovery center (this could be at the county level)	National Government, Kilifi County Government and NGOs, National Government, Kilifi County Government, National Police Service, NGOs, GVRC, COVAW, HAK	1,000,000 for consultancy services Construction of GBV recovery center subject to a detailed Bill of Quantities
Assess the existing early warning systems to produce and transmit reliable and timely information and forecasts for the implementation of GBV anticipatory actions ahead of climate hazards	Strengthen gender-responsive governance-inclusion of women, youth and PADs through capacity building across Dzikunze, Midugani, Mwele and Ndugumnani villages		Kilifi County Government and NGOs	5,000 per person per day for training
Use of existing women associations such as VSLAs to train women and girls on their rights and GBV response strategies, helplines and recovery centers for women and girls as well as the role of	Procure/provide GBV response mechanisms such as transfer vehicles to health facilities and specialized first aid and rape kits to treat and preserve evidence for prosecution respectively		Kilifi County Government, Civil Society, NGOs, Private Sector	30,000 per VSLA BOQ for GBV response mechanisms

Adaptation actions			Potential implementing partners	Budget (KES)
Short-term	*Medium*	*Longterm*		
men in peaceful conflict resolution				
Organize and hold youth boot camps on guidance and counselling focusing on drugs, antisocial behavior and reproductive health	Promote vaccination of children and youth against diseases such as cholera which could arise and spread on account of climate hazards Support school feeding programmes and scholarships	Construct drug rehabilitation centre targeting the youth (this can be done at the county level)	National Government, NACADA, Kilifi County Government, NGOs, Private Sector	100,000- Boot camp, Vaccination at local medical facilities, BOQ required for drug rehabilitation centre School feeding and scholarships needs assessment
Facilitate PADs to link with organizations that can provide infrastructural assistance such as wheelchairs, specialized training and entrepreneur skills			Kilifi County Government, NGOs, Private Sector	50,000 for meetings

5.4 Resource Mobilisation

Implementation of the Ward Adaptation Plan will rely on a diversified mix of financial, human, and technical resources drawn from county, national, donor, and community-based funding streams. The primary sources include county budget allocations, where relevant departments such as Water, Environment and Natural Resources, Agriculture, Livestock and Blue Economy, and Public Works and Infrastructure will integrate adaptation priorities into their Annual Development Plans and sector budgets. Sokoke Ward is projected to receive at least KES 2–3 million annually earmarked for ward-level climate resilience, as required under the Climate Change Act, 2023.Additional financing will be accessed through the Kilifi County Climate Change Fund (CCCF), which channels grants from development partners including the World Bank's FLLoCA programme, supporting locally led adaptation initiatives. At the national level, Sokoke Ward will benefit from funds such as the National Drought Emergency Fund and other climate financing mechanisms including the Kenya Adaptation Fund Programme, coordinated through the National Treasury's Climate Finance Unit and implemented in collaboration with agencies like NDMA and KMD.

Development partners and NGOs, including Plan International, WWF, World Vision, Kenya Cereal Enhancement Programme (KCEP), and Botanical Garden Conservation International (BGCI), will continue to support ecosystem restoration, capacity building, and livelihood diversification projects. Community contributions through in-kind labor, local materials, savings groups, Water Resource Users Associations (WRUAs), and small user fees will foster ownership and sustainability. The private sector will be engaged through partnerships for water infrastructure development, crop and livestock insurance, microfinance for smallholder irrigation, digital early warning SMS systems, and corporate social responsibility initiatives. Overall, Sokoke Ward expects to mobilize approximately KES 25–35 million over five years from this blended financing framework, ensuring continuity and long-term sustainability of adaptation interventions.

6 Chapter 6: Adaptation Monitoring and Evaluation

6.1 Introduction

Monitoring, Evaluation, and Learning (MEL) are critical components of the Ward Adaptation Plan to ensure that the proposed actions are achieving their intended outcomes and to facilitate continuous improvement. This chapter outlines how Sokoke Ward will track progress of adaptation interventions (Monitoring), assess their effectiveness and impact (Evaluation), and capture lessons to inform future climate-resilience planning (Learning). A robust M&E framework not only provides accountability to the community and funders but also enables adaptive management, meaning the Ward Climate Committee can adjust strategies based on what is working or not working on the ground. Given the long-term nature of climate change, M&E will help maintain momentum and demonstrate how resilience is being built incrementally.

The approach will utilize participatory methods so that community members are involved in monitoring their own resilience, alongside technical indicators aligned with county and national reporting requirements. The Climate Change Act 2023 reinforces the importance of reporting and transparency in climate actions, and this plan's M&E system will align with those national guidelines while tailoring to local context.

6.2 The Adaptation Theory of Change

The theory of change is a conceptual model that explains how the planned interventions will lead to the desired climate resilience outcomes. In simplified form:

Inputs/Activities:

- Construction of water pans, earth dams, and solar-powered boreholes.
- Rainwater harvesting systems at household and community levels.
- Training on climate-smart and regenerative agriculture.
- Livelihood diversification (beekeeping, poultry, fish farming, tailoring, small businesses).
- Tree planting and forest restoration, soil and water conservation, alternative energy promotion.
- Fencing farms, establishing wildlife water points, planting deterrents, and awareness campaigns.
- Infrastructure improvements (footbridges, drainage, wind-resistant housing) and hazard preparedness training

Outputs:

- Operational water systems, trained farmers adopting new practices, households with food storage, livelihood activities initiated.
- Trees planted and surviving, forest and soil conservation measures functional, environmental awareness increased.
- Farms protected from wildlife, households practicing coexistence, and reduced damage from floods and whirlwinds.

Outcomes:

- Increased water access during dry seasons, adoption of drought-tolerant crops and livestock, reduced crop losses, improved food security, diversified incomes.
- Improved forest and soil health, reduced unsustainable resource use, enhanced community participation in natural resource management.
- Reduced human-wildlife conflicts, decreased flood and whirlwind damage, improved household and community preparedness.

- **Impact (Goal):** A climate-resilient Sokoke Ward, with secure livelihoods, enhanced adaptive capacity, and reduced vulnerability to droughts, floods, pests, and whirlwinds

The theory of change for Sokoke can be summarized as: "IF we provide reliable water sources, promote resilient farming and livelihoods, strengthen natural resource management, enhance environmental restoration and enhance local institutions' capacity, it is expected the community's exposure and sensitivity to climate hazards will decrease and their adaptive capacity will increase, because they will have the necessary resources, knowledge, and organization to cope with climate stresses." This will lead to the ultimate impact of a more climate-resilient Pongwe Kikoneni Ward with secure livelihoods and wellbeing despite climate change.

This logical flow will guide M&E. Each level (input -> output -> outcome -> impact) will have specific indicators to check. For instance, training farmers (input) should lead to output of farmers implementing new techniques, which yields outcome of yield stability in drought, contributing to impact of overall food security. If a link is failing.

6.3 Performance Indicators for Adaptation

Based on the theory of change, a set of Performance Indicators will be used to measure progress and effectiveness. These indicators will be both quantitative and qualitative, and they align with national adaptation M&E frameworks where possible (Kenya's NAP/NCCAP indicators) as well as community-defined measures of success. The performance indicators are structured at various levels:

Input/Output Indicators:

1. WATER SECURITY AND MANAGEMENT
   - Number of household rainwater harvesting systems installed (gutters + tanks).
   - Number of schools equipped with high-capacity rainwater harvesting systems.
   - Number of water pans constructed.
   - Number of manual and solar-powered pumps installed on existing wells.
   - Number of women, youth, and PADs using solarised/manual hand pumps.
   - Number and percentage of women, youth, and PADs in water management committees.
   - Amount of budget allocated and utilised for water infrastructure.
   - Number of boreholes drilled, equipped, and solarised.
   - Number of boreholes fully operational and supplying water year-round.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOOD
   - Number of farmers trained on Resilience Design techniques.
   - Number of farmers trained on CSA practices (land prep, planting, pest control, post-harvest).
   - Acreage under drought-tolerant crops supported with certified seeds.
   - Number of Farmer Field Schools established and functional.
   - Number of demonstration farms established by women and youth lead farmers.
   - Number of farmers receiving inputs (certified seeds, fertilizers, pesticides).
   - Number of households supported with organic manure production and composting.
   - Number of farmers adopting terracing, zai pits, swales, and other soil/water conservation measures.
   - Number of livestock fodder seed banks and pasture plots established.
   - Number of households receiving improved livestock breeds (Gala goats, dairy cows).
   - Number of farmers sensitised on crop/livestock insurance.

- Number of households supported with grain storage equipment (hermetic bags, grain banks).
- Number of village pest and disease surveillance teams established and trained.
- Number of households receiving indigenous/fruit seedlings.
- Number of community seedling distribution events held.
- Number of on-farm tree planting sessions conducted.
- Number of acres of mangrove restored through enrichment planting.
- Number of community trainings on climate change and environmental conservation.
- Number of environmental conservation awareness sessions held.
- Number of households adopting sustainable charcoal alternatives or clean energy.
- Budget allocated and utilised for restoration activities.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Number of women and youth trained in alternative livelihoods (beekeeping, poultry).
- Number of modern beehives, incubators, or eco-tourism boats distributed.
- Number of households supported with livelihood start-up kits.
- Number of community groups formed for value addition and market linkages.
- Number of training sessions on business development, value addition, and financial literacy.
- Number of social protection initiatives implemented (food distribution, cash transfer mobilizations).
- Number of school feeding programmes supported.
- Budget allocations for livelihood diversification implemented.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Number of villages with functional flood Early Warning Systems (EWS) linked to KMD.
- Number of community emergency response teams trained.
- Number of community emergency response teams equipped (boats, PPE, first-aid kits).
- Number of health facility constructed.
- Number of NBS windbreaks (Neem, Casuarina) planted across all villages

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Number of community sensitization sessions on human–wildlife conflict hotspots.
- Number of households supported with HWC reporting mechanisms (hotline awareness).
- Number of chili cultivated farms
- Number of reported HWC cases logged and submitted to KWS for follow-up.
- Funding allocated and utilised for HWC mitigation (fencing, awareness, surveillance).

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Number of gender-responsive governance trainings conducted.
- Number and percentage of women, youth, and PADs participating in ward committees.
- Number of VSLAs supported with GBV awareness and rights training.
- Number of GBV anticipatory action sessions conducted for drought, floods, and strong winds.
- Number of GBV response kits (first-aid kits, rape kits) distributed to health facilities.
- Number of early warning systems integrating GBV alert messaging.
- Number of PADs supported with assistive devices, rehabilitation linkages, or livelihood equipment.

- Number of youth bootcamps conducted on drugs, behaviour change, and reproductive health.
- Number of awareness campaigns on GBV helplines and safe reporting mechanisms.
- Number of meetings with PAD support organizations facilitated.

Outcome Indicators:

1. WATER SECURITY AND MANAGEMENT

- Reduction in average time spent collecting water (minutes/day), disaggregated by women, youth, and PADs.
- Increase in household water storage capacity (litres/HH).
- Percentage of water facilities functional year-round.
- Increase in number/percentage of women, youth, and PADs with independent access to water sources (boreholes, solarised pumps, kiosks).
- Increased reliability of water supply during drought periods (days/months of service continuity).

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOOD

- Stability of crop yields over 3–5 seasons among supported farmers.
- Improvement in soil health indicators, including:
- Increase in soil organic matter (%)
- Increase in soil moisture retention capacity
- Improvement in soil nutrient status (N, P, K)
- Improved aggregate stability and soil structure
- Percentage of farmers adopting integrated soil and water conservation (terracing, zai pits, swales).
- Increase in acreage under diversified and climate-resilient farm systems.
- Increase in household-level adoption of post-harvest management technologies (grain banks, hermetic bags).
- Percentage of women, youth, and PADs participating in CSA and Resilience Design training cycles.
- Increase in on-farm and community tree cover (%) over baseline.
- Reduction in incidents of illegal harvesting and charcoal burning.
- Increased community participation in environmental conservation and sustainable harvesting practices.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Percentage increase in household income from diversified livelihoods (eco-tourism, poultry, beekeeping).
- Increase in the number of households with multiple income streams across seasons.
- Percentage of women, youth, and PADs accessing new livelihood opportunities.
- Increased financial literacy and business skills among trained groups.
- Increase in households benefiting from school feeding, cash transfers, or food-for-work schemes.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Reduction in households exposed to flood-prone or erosion hotspots.
- Percentage of villages receiving timely Early Warning System (EWS) alerts.

- Community response rate to EWS alerts (attendance at assembly points, evacuation compliance).
- Reduction in flood-related damages to dwellings and access roads.
- Level of preparedness and readiness of community emergency response teams (measured via drills).

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Reduction in frequency of human–wildlife conflict incidents (elephants attacks).
- Increase in the number of households protected by solar-powered fences.
- Increase in community capacity to report and respond to HWC events (via KWS hotline).
- Increased community compliance with safe-behaviour protocols in HWC hotspots.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Increase in the number and proportion of women, youth, and PADs participating in ward-level governance and adaptation committees.
- Reduction in drudgery for women, youth, and PADs due to improved water access and resilient infrastructure.
- Percentage increase in women's and PADs' access to credit, extension services, and productive resources.
- Increase in women's ownership of productive assets (livestock, equipment).
- Improvement in community knowledge and awareness of GBV risk mitigation during climate hazards.
- Increased utilisation of GBV response services (safe shelters, first-aid kits, referral pathways).

Impact (Goal) Indicators:

1. WATER SECURITY AND MANAGEMENT

- Percentage of the ward population with year-round access to safe and reliable water.
- Increase in average litres of water available per person per day across villages.
- Long-term reduction in water scarcity–related vulnerabilities for women, youth, and PADs.
- Sustained decrease in waterborne diseases due to improved water quality and availability.
- Community-wide resilience to prolonged drought through diversified water sources and storage systems.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOOD

- Sustained increase in crop yields above baseline across 3–5 year periods.
- Long-term reduction in seasonal yield fluctuations linked to droughts, floods, and pest outbreaks.
- Measurable improvements in soil health at landscape level, including:
- Higher soil organic carbon (SOC)
- Increased vegetation cover (NDVI index)
- Reduced erosion and sediment loss
- Enhanced household food security across all seasons.
- Increased biodiversity within integrated farming systems (trees, fodder, pollinator presence).

- Increases in forest cover and on-farm trees compared to baseline.
- Restoration of degraded landscapes evidenced by improvements in NDVI and tree density.
- Reduction in illegal tree cutting, and charcoal burning.
- Enhanced ecosystem services, including soil stability, and microclimate regulation.
- Long-term improvement in habitat quality for biodiversity.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Sustained increases in household incomes from diversified livelihoods (beekeeping, eco-tourism, poultry).
- Improved household resilience to income shocks from climate hazards.
- Decreased poverty levels due to stable and multi-stream livelihoods.
- Increased school attendance and nutritional outcomes for children due to social protection schemes.
- Greater financial stability among women, youth, and PAD households.

4. RESILIENT INFRASTRUCTURE AND HUMAN SETTLEMENTS

- Significant reduction in climate-related damage to homes, roads, and community facilities.
- Communities demonstrating effective disaster preparedness and response during floods and storms.
- Long-term reduction in displacement, injuries, and asset losses during climate hazards.
- Increased structural safety and durability of settlements in strong-wind and flood-prone areas.

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Long-term reduction in human injuries, livestock losses, and crop destruction from wildlife.
- Improved coexistence between communities and wildlife due to effective mitigation (fencing, reporting systems).
- Reduced economic losses from HWC-related disruptions.
- Increased safety and mobility for women, children, and PADs in high-risk zones.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Percentage increase in women's, youth's, and PADs' participation in climate governance and local decision-making structures.
- Strengthened continuity of SRHR, maternal health, and GBV response services during climate hazards.
- Increased leadership roles held by women, youth, and PADs in adaptation implementation.
- Growth in the number of marginalised and vulnerable groups leading or influencing climate adaptation actions.
- Reduction in GBV risks during droughts, floods, and strong winds due to anticipatory action systems.

6.4 Adaptation Reporting and Learning

Reporting: Sokoke Ward will establish a clear reporting system to ensure information flows to all stakeholders i.e., community members, county authorities, and funding partners. Internally,

the WCCPC will prepare quarterly progress reports that document activities done, outputs achieved, challenges encountered, and expenditures made. These will be shared in ward public barazas for transparency (so the community hears what's been done and can give feedback) and submitted to the County Climate Change Unit. Annually, a more comprehensive Ward Adaptation Report will be produced, summarizing the year's progress against the plan's targets and indicators. This annual report can feed into the county's climate change reporting which is required by the Climate Change Act, ensuring Sokoke Ward's experiences contribute to county and national adaptation tracking.

The annual report will highlight case studies or success stories (e.g., "Farmer groups in Ndungumnani doubled their harvest despite the 2025 drought by using new techniques a success attributed to our adaptation training"). It will also not shy away from discussing failures or shortfalls, as these provide learning.

At community level, participatory monitoring meetings will be held. For instance, after each rainy season, community members can gather to discuss: did the drainage improvements prevent flooding? Did the new seeds perform better in this season? Such qualitative feedback will be recorded and included in reports.

The Ward Administrator and WCCPC Chair will also report in county forums (e.g., County Adaptation Planning Committee meetings) on progress, championing their ward and possibly leveraging more support by showcasing results.

Learning and Reflective Review: Adaptation is a long journey with uncertainty, so fostering a culture of learning is vital. Twice during the implementation period (e.g., mid-term in year 3, and end-term in year 5), the ward will conduct an outcome evaluation and learning workshop. This involves bringing together community representatives, implementing partners, and possibly external evaluators to systematically review what outcomes have been achieved and why. Key questions: Which interventions are most effective? Are there unintended impacts? What obstacles are reducing effectiveness (maybe maintenance issues, or cultural barriers to adoption)? For example, maybe the uptake of an improved cookstove program is low – through reflection the group might learn women prefer a different design, or fuel is an issue, leading to adjusting that action. These learning workshops will use tools like community scorecards, ranking of interventions by community benefit, and problem tree analysis for any persistent challenges (similar to the PCVA style but now focusing on project performance). Lessons learned will be documented and used to refine the adaptation plan. The Ward Climate Committee is empowered to tweak or reprioritize actions based on evidence. For instance, if a particular crop variety fails, they will note it and try an alternative next season.

Knowledge Sharing: Sokoke Ward will also engage in horizontal learning i.e., exchanging experiences with other wards or counties. If something worked really well here (say the VSLA approach to fund water tanks), we can host visitors from nearby wards to showcase it, and similarly we can learn from others (maybe a neighbouring ward piloted a solar irrigation scheme that we could replicate). The county climate change unit can facilitate such inter-ward learning exchange visits or meetings.

Feedback to Policy: Through reporting channels, the ward's experiences may also inform higher level policies. For example, if the ward finds that insurance uptake is low because of certain policy constraints, this feedback can go to the national level via NDMA or the National Climate Change Council to refine such programs. The 2023 Act calls for progressive improvement of climate response, our ward's data and stories become part of that national learning loop.

Sustainability of M&E: The Ward Committee will assign an M&E focal person (or sub-committee) to drive these activities. But monitoring won't be just "external evaluation"; it will be integrated into community life e.g., water committees will track water usage and downtime of boreholes as part of their routine, farmer groups will collectively observe and record crop performance. By embedding monitoring tasks into existing structures, we ensure continuity even if specific projects end.

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9 Annexure

1. Priority tree and fruit species

Scientific Name	Common Name
Azadirachta indica	Neem
Afzelia quanzensis	Pod Mahogany / Mkongo
Casuarina equisetifolia	Whistling Pine / Casuarina
Milicia excelsa	African Teak / Iroko / Mvule
Gmelina arborea	Gmelina / White Teak
Tectona grandis	Teak
Grevillea robusta	Silk Oak / Grevillea
Senna spp.	Senna / Cassia
Moringa oleifera	Moringa / Drumstick Tree
Azalea spp.	Azalea
Citrus sinensis	Sweet Orange
Anacardium occidentale	Cashew
Mangifera indica	Mango
Cocos nucifera	Coconut
Carica papaya	Papaya